

# From Forecasts to Batteries: Grouped Decision-Focused Model Averaging for Capacity Commitment and Storage Siting on Five Continents

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## Abstract

Grid operators must commit capacity a day ahead, and planners must decide where to install new batteries. Both decisions rest on how the many available demand forecasts—operators’ own forecasts, statistical models, machine learning, models driven by archived numerical weather forecasts, and pretrained foundation models—are combined. We propose *grouped decision-focused model averaging* (GDMA), which jointly learns a small number of latent segments of decision units and segment-specific combination weights by minimising the realised decision cost, with each unit’s own cost asymmetry, rather than forecast error; a soft version shrinks each unit’s weights towards its segment, and a fair version counts every unit’s relative improvement equally. We prove a finite-sample oracle inequality that separates the cost of learning segment memberships from the cost of learning weights, show that GDMA is asymptotically optimal and attains the risk of the infeasible unit-specific optimum when segments exist, and prove that it recovers the segments with probability approaching one. A planning stage turns the residual risk of the learned rule into an efficient or a fair siting of storage. On 125 grid areas in North America, Europe, Oceania, Asia and Africa, using public hourly data from 2018 to 2026 and only information available at the day-ahead gate closure, soft GDMA lowers the stylised out-of-sample commitment cost by 12% worldwide relative to a reference forecaster (the operators’ official forecast in the United States, a model forecast elsewhere) and by 4 percentage points relative to each area’s best single candidate; for the 46 U.S. balancing authorities the saving relative to their official forecasts is 13–17%. The gain comes from combining forecasts for the decision rather than for accuracy and from sharing structure across areas; the grouped variants, two-step clustering and plain GDMA are statistically indistinguishable in a model confidence set, and decision-focused shrinkage comes close. Forecast-then-commit and an unconstrained quantile-regression combination do worse, a decision-focused neural mixture-of-experts gate is 7 percentage points more expensive, and the simplex-constrained rules avoid the extreme weeks that ruin the other rules. Soft GDMA makes 6% of grid areas worse off than their own best candidate and its fair version 3%; we find no systematic income gradient, though the data cannot exclude a moderate one. Archived deterministic and ensemble weather forecasts halve the gap to the operators’ own forecasts during Winter Storm Elliott, from 20% to about 10%, but do not close it. In a stylised storage model, the reference rule needs 34 GW of optimally sited batteries to match the commitment cost of soft GDMA without any; siting storage on the error profile of the wrong rule loses 2–4% of a programme’s value, and maximin-fair

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siting raises the worst-served continent’s benefit at a price of fairness of 7–9%.

*Keywords:* model averaging, decision-focused learning, latent groups, foundation models, fairness, capacity commitment, battery storage siting, energy management

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## 1. Introduction

Electricity systems are managed through many local decision units that commit resources before uncertainty resolves. Every day, each of the balancing authorities, national system operators and regional market operators that run the world’s grids must commit enough capacity to meet the next day’s hourly demand. Committing too little forces expensive real-time purchases, the activation of reserves or, in the worst case, load shedding; committing too much means paying for capacity that sits idle. The cost of a shortfall is not the same everywhere: an operator embedded in a dense web of interconnections can import from its neighbours, while an isolated system cannot. Winter Storm Uri in February 2021 is the extreme example: failures of frozen generators and gas supply forced the weakly interconnected Texas grid to shed load for days, and its limited ability to import deepened the crisis, while neighbouring systems with more import capability shed much less (Busby et al., 2021; FERC, NERC and Regional Entity Staff, 2021). At the same time, governments are committing to large battery-storage programmes, and planners must decide where those batteries should go.

Operators and planners do not lack forecasts. Operators publish their own day-ahead forecasts, simple persistence and calendar rules are always available, statistical and machine-learning models can be trained on the data, and pretrained foundation models now forecast any series zero-shot (Ansari et al., 2024). None dominates everywhere, which makes combining them attractive. How to combine them is less obvious. The forecast-combination and model-averaging literature, from Bates and Granger (1969) to the asymptotic theory of Hansen (2007), Hansen and Racine (2012), Zhang et al. (2013) and Zhang and Liu (2023), offers two options when there are many series. Weights can be estimated series by series, which respects heterogeneity but produces noisy weights when histories are short—the root of the “forecast-combination puzzle” that simple averages often beat estimated weights (Smith and Wallis, 2009; Claeskens et al., 2016). Or one set of weights can be estimated for all series, which is precise but imposes the same rule on systems that face different conditions. Deep learning offers a third option, a network that maps features to weights (Montero-Manso et al., 2020; Gong et al., 2026), at the price of thousands of parameters and weights that are hard to audit. All three typically judge a combination by its statistical accuracy, although the planner is judged by the cost of the resulting decisions; the decision-focused literature in operations management has shown that learning directly for the decision can be much better than forecasting accurately and then optimising (Ban and Rudin, 2019; Bertsimas and Kallus, 2020; Elmachoub and Grigas, 2022).

This paper proposes *grouped decision-focused model averaging* (GDMA) and uses it to answer three management questions: how should grid operators on five continents combine the forecasts they have into day-ahead capacity commitments, who gains and who loses from a common planning rule, and where should new batteries be installed? GDMA assumes that decision units fall into a small number of latent segments whose members are best served by the same combination of candidate rules, and estimates the segments and the segment-specific weights jointly by minimising the realised system-wide decision cost, in money, with each unit’s own cost asymmetry. A soft version shrinks each unit’s

own weights towards those of its segment. The number of segments and the shrinkage intensity are chosen by a time-ordered hold-out. Pooled weights, unit-specific weights and shrinkage towards pooled weights are special cases, and the combined commitment is always a convex combination of commitments the operator already understands. A second, planning stage takes the residual shortfall risk left by the learned rule and allocates a storage budget across grid areas.

We make five contributions.

1. **Method.** We formulate the joint learning of latent segments and combination weights under an asymmetric, unit-specific decision loss, give a  $k$ -means-type algorithm with regret-space seeding in which every step is a small convex program, and link it to a storage-siting problem that is solved exactly by a greedy algorithm (Section 3). To our knowledge, neither the model-averaging, the decision-focused learning nor the latent-group panel literature (Bonhomme and Manresa, 2015; Su et al., 2016) has proposed this estimator.
2. **Learning theory.** We prove a finite-sample oracle inequality that separates the cost of learning memberships ( $\log G/T$  per unit) from the cost of learning weights ( $GM \log(NT)/(NT)$ ), which shows how the cost of grouping scales relative to unit-specific weights when histories are short; we show that GDMA is asymptotically optimal and, under a segment structure, attains the risk of the infeasible benchmark in which every unit knows its own optimal weights; that it recovers the segments with probability approaching one; and that on this event it coincides exactly with an estimator told the segments (Section 4).
3. **Evidence on five continents.** We evaluate all rules with the information available at the day-ahead gate closure. On 46 U.S. balancing authorities with their official forecasts (Section 6) and on a global panel of 125 grid areas in North America, Europe, Oceania, Asia and Africa (Section 7), soft GDMA lowers the stylised out-of-sample commitment cost by 13–17% relative to the official forecasts in the United States and by 12% worldwide relative to a reference forecaster, beats combining the same forecasts for accuracy by 2–3 points in the United States (at  $W = 28$ ) and 8–10 points worldwide, and beats a decision-focused neural mixture-of-experts gate by seven percentage points worldwide. Decision-focused shrinkage and two-step clustering come close, and a model confidence set cannot separate the grouped variants from one another, so the evidence is for decision-focused combination with cross-sectional structure rather than for grouping alone. A pretrained foundation model and models driven by archived day-ahead deterministic and ensemble weather forecasts enter the candidate set on equal terms; the ensemble information roughly halves, but does not close, the gap to the operators’ own forecasts during Winter Storm Elliott, the most abrupt event in the sample.
4. **Fairness.** We audit how the gains are distributed across grid areas and propose a fair variant that counts every area’s relative improvement equally. Soft GDMA makes 6% of areas worse off than their own best candidate, against 24% for pooled weights and 42% for the neural gate, shows no systematic gradient with national or state income that the data can detect, and its fair variant halves the share of areas made worse off at no cost in total (Section 7.4).
5. **Storage siting.** In a stylised storage model, better commitment rules reduce what storage can add: the reference rule needs 34 GW of optimally sited batteries to reach the cost that soft GDMA achieves without any. The optimal location of a storage programme depends on the commitment rule, but siting on the wrong rule’s error

profile loses only 2–4% of a programme’s value, and maximin-fair siting helps the worst-served continent at a price of fairness of 7–9% (Section 8).

The paper speaks to the theme of this special issue in both directions it describes: it develops a statistical learning method with guarantees, motivated by a concrete management problem, and it uses new, openly available data and models to answer a planning question of continental scale. Replication code and data-construction scripts are available in a public repository (see the data availability statement).

## 2. Related literature and the research gap

*Model averaging and forecast combination..* Since [Bates and Granger \(1969\)](#), combining forecasts has been one of the most reliable ways to improve predictive accuracy; [Timmermann \(2006\)](#) and [Wang et al. \(2023\)](#) review five decades of work. Frequentist model averaging chooses weights by minimising an estimate of risk—a Mallows criterion ([Hansen, 2007](#)), leave-one-out or  $K$ -fold cross-validation ([Hansen and Racine, 2012](#); [Zhang and Liu, 2023](#)), or forward validation for time series ([Zhang and Zhang, 2023](#))—and establishes asymptotic optimality, i.e. that the chosen weights attain the lowest attainable risk. The framework now covers dependent data ([Zhang et al., 2013](#)), quantile losses ([Lu and Su, 2015](#)), general losses ([Yu et al., 2024](#)), time-varying weights ([Sun et al., 2021](#)), covariate-dependent weights ([Gong et al., 2026](#)), covariate shift ([Zhang et al., 2026](#)) and prediction intervals ([Qu et al., 2025](#)). The forecast-combination puzzle—that equal weights frequently beat estimated weights—is attributed mainly to estimation error in the weights ([Smith and Wallis, 2009](#); [Claeskens et al., 2016](#)); remedies shrink estimated weights towards equal weights ([Diebold and Shin, 2019](#)) or restrict them ([Zou et al., 2025](#)). When many series must be forecast, weights are either estimated series by series or learned from series features (FFORMA, [Montero-Manso et al., 2020](#)); [Aiolfi and Timmermann \(2006\)](#) cluster *forecasting models* by past performance. All of these approaches either treat each target series separately or impose one rule on all of them, and none groups the *target units*.

*Decision-focused learning..* Operations research has long recognised that the loss that matters is the decision cost ([Granger, 1969](#)). The data-driven newsvendor literature learns order quantities directly from data ([Levi et al., 2015](#); [Ban and Rudin, 2019](#)); prescriptive analytics maps covariates to decisions ([Bertsimas and Kallus, 2020](#)); and “smart predict-then-optimize” trains predictors on decision regret ([Elmachtoub and Grigas, 2022](#)); see [Mandi et al. \(2024\)](#) for a survey. In energy forecasting, [Wang et al. \(2019\)](#) combine probabilistic load forecasts by minimising pinball loss on the simplex, one series at a time. Most of this literature trains single models or combines forecasts for one unit. An important exception is data pooling ([Gupta and Kallus, 2022](#)), which shrinks the solutions of many small newsvendor-type problems towards a common anchor and shows that pooling pays even when the problems are unrelated; decision-focused shrinkage of combination weights towards pooled weights, one of our benchmarks and the  $G = 1$  case of soft GDMA, applies the same idea to forecast combination.

*Deep learning for forecasting and for combination weights..* Pretrained time-series foundation models such as Chronos ([Ansari et al., 2024](#)) now forecast unseen series zero-shot with accuracy close to models trained on the target data, and deep networks are increasingly used to *produce* combination weights: FFORMA maps series features to weights

with gradient boosting (Montero-Manso et al., 2020), and mixture-of-experts model averaging uses a softmax neural gate with covariate-dependent weights (Gong et al., 2026). These approaches are flexible, but they fit thousands of parameters, their weights are hard to audit, and—except for Gong et al. (2026), who study conditional generative models—they come without optimality guarantees for the combined decision. We include both a foundation model (as a candidate) and a decision-focused neural gate (as a competitor) in our application.

*Storage siting.* Where to place grid-scale batteries is usually decided with engineering models of the transmission network, solved by decomposition or stochastic programming (Piansky et al., 2024; Larroux et al., 2026). These models take the uncertainty that storage must absorb as an input. We make that input endogenous: the residual commitment risk depends on how operators combine their forecasts, so the value and the best location of storage depend on the forecasting policy.

*Latent group structures.* Panel econometrics models unobserved heterogeneity through latent groups estimated by  $k$ -means-type algorithms (Lin and Ng, 2012; Bonhomme and Manresa, 2015; Ando and Bai, 2016) or classifier-Lasso (Su et al., 2016). Gu and Volgushev (2019) extend latent groups to panel quantile regression, which is the closest statistical relative of GDMA: a grouped quantile regression of demand on the candidate forecasts would be a grouped, unconstrained version of our quantile-regression benchmark. These methods group regression parameters, typically fixed effects, for inference on the parameters themselves; GDMA groups simplex weights over the outputs of existing forecasting systems, with unit-specific critical ratios, and is judged by the out-of-sample cost of the resulting decisions.

*The gap.* Table 1 summarises the position of GDMA. No existing method simultaneously (a) combines a library of existing candidate rules, including machine-learning and foundation-model forecasts, (b) chooses weights by the downstream decision cost with unit-specific cost asymmetries, and (c) shares statistical strength across many decision units through latent segments whose number is chosen from data, with guarantees on optimality and segment recovery; nor does any link the resulting operational risk to the siting of new storage. The individual ingredients exist—decision-focused combination, shrinkage across problems, latent groups—and the contribution is their combination, the theory for it, and its use for a planning decision; as the applications show, much of the gain over the best single forecaster is already delivered by decision-focused shrinkage, and grouping adds a smaller refinement. Filling this gap matters most when windows are short—because systems change and old data are stale—and units are many, which is exactly the situation of system-wide infrastructure planning.

### 3. Grouped decision-focused model averaging

#### 3.1. The system-wide commitment problem

A planner is responsible for  $N$  decision units (in our application the balancing authorities of the U.S. grid). In period  $t$  unit  $i$  must commit a quantity  $q_{it}$  before demand  $y_{it}$  is realised. Committing too little costs  $u_i > 0$  per unit of shortfall (emergency purchases, reserve activation or, in the limit, curtailed load); committing too much costs  $o_i > 0$  per unit of excess (idle capacity that was paid for). The period cost is the newsvendor cost

$$C_i(q, y) = u_i (y - q)^+ + o_i (q - y)^+ = (u_i + o_i) \rho_{\tau_i}(y - q), \quad \tau_i = \frac{u_i}{u_i + o_i}, \quad (1)$$

Table 1: Positioning of GDMA relative to existing approaches.

| Approach                                          | Combines<br>candidates | Decision-<br>focused loss | Shares strength<br>across units | Heterogeneous<br>weights |
|---------------------------------------------------|------------------------|---------------------------|---------------------------------|--------------------------|
| Per-series combination / JMA / forward validation | ✓                      |                           |                                 | ✓                        |
| Pooled combination                                | ✓                      |                           | ✓                               |                          |
| Quantile / general-loss model averaging           | ✓                      | ✓                         |                                 | ✓                        |
| Shrinkage towards equal or pooled weights         | ✓                      |                           | partial                         | ✓                        |
| Data pooling in stochastic optimisation           |                        | ✓                         | partial                         | ✓                        |
| Feature-based weights (FFORMA)                    | ✓                      |                           | ✓                               | ✓                        |
| Neural mixture-of-experts gate                    | ✓                      | (✓)                       | ✓                               | ✓                        |
| Data-driven newsvendor, SPO                       |                        | ✓                         | ✓                               |                          |
| Latent-group panel (quantile) models              |                        | (✓)                       | ✓                               | ✓                        |
| <b>GDMA (this paper)</b>                          | ✓                      | ✓                         | ✓                               | ✓                        |

where  $\rho_\tau(r) = r(\tau - \mathbb{1}\{r < 0\})$  is the check function. The system cost that the planner cares about is  $\sum_i \sum_t C_i(q_{it}, y_{it})$ ; under (1) the cost-minimising commitment is the conditional  $\tau_i$ -quantile of demand (Gneiting, 2011). Critical ratios  $\tau_i$  differ across units because the consequences of a shortfall differ: an operator that is tightly interconnected with its neighbours can import at moderate cost, while an isolated one cannot.

For every unit a library of  $M$  candidate commitment rules is available. Candidate  $m$  proposes  $q_{it}^{(m)}$ , computed from information available before period  $t$ ; we write  $\mathbf{q}_{it} = (q_{it}^{(1)}, \dots, q_{it}^{(M)})^\top$ . The candidates may come from any source: an operator’s own published forecast, simple persistence rules, statistical models or machine-learning models, each turned into a commitment by adding its own  $\tau_i$ -quantile residual margin. The planner combines them linearly,

$$q_{it}(\mathbf{w}) = \mathbf{q}_{it}^\top \mathbf{w}, \quad \mathbf{w} \in \Delta_M = \left\{ \mathbf{w} \in \mathbb{R}^M : w_m \geq 0, \sum_m w_m = 1 \right\}.$$

The simplex constraint keeps the combined commitment inside the range spanned by the candidates, which is what makes the combination interpretable and auditable by operators, and it is known to stabilise estimated weights (Zou et al., 2025).

### 3.2. Three ways to estimate the weights

Let the estimation window contain periods  $t = 1, \dots, T$  for which both candidate commitments and realised demand are observed (a forward-validation sample in the terminology of Zhang and Zhang, 2023). Write  $\hat{R}_i(\mathbf{w}) = T^{-1} \sum_{t=1}^T C_i(\mathbf{q}_{it}^\top \mathbf{w}, y_{it})$  for the empirical cost of unit  $i$ .

*Unit-specific weights.*  $\hat{\mathbf{w}}_i = \arg \min_{\mathbf{w} \in \Delta_M} \hat{R}_i(\mathbf{w})$ . This respects heterogeneity but estimates  $M - 1$  free parameters from  $T$  serially dependent observations per unit; it is the decision-focused counterpart of per-series forecast combination, and suffers from the same estimation noise that underlies the forecast-combination puzzle (Smith and Wallis, 2009; Claeskens et al., 2016).

*Pooled weights.*  $\hat{\mathbf{w}} = \arg \min_{\mathbf{w}} \sum_i \hat{R}_i(\mathbf{w})$ . This uses all  $NT$  observations but forces a single policy on units that may need different ones.

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**Algorithm 1** GDMA for a fixed number of segments  $G$ 

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**Require:** candidate commitments  $\{\mathbf{q}_{it}\}$ , demand  $\{y_{it}\}$ , costs  $\{u_i, o_i\}$ ,  $G$ , number of starts  $S$

- 1: compute unit-specific weights  $\widehat{\mathbf{w}}_i$  and the regret matrix  $D_{ij} = \widehat{R}_i(\widehat{\mathbf{w}}_j) - \widehat{R}_i(\widehat{\mathbf{w}}_i)$
- 2: **for**  $s = 1, \dots, S$  **do**
- 3:   draw seeds  $j_1, \dots, j_G$  by  $k$ -means++ on  $D$  and set  $\mathbf{w}_g \leftarrow \widehat{\mathbf{w}}_{j_g}$
- 4:   **repeat**
- 5:      $\gamma_i \leftarrow \arg \min_g \widehat{R}_i(\mathbf{w}_g)$  for all  $i$  ▷ assignment step
- 6:     re-seed any empty segment with the unit of largest regret
- 7:      $\mathbf{w}_g \leftarrow \arg \min_{\mathbf{w} \in \Delta_M} \sum_{i: \gamma_i = g} \widehat{R}_i(\mathbf{w})$  for all  $g$  ▷ pooled convex step
- 8:   **until** memberships do not change
- 9: **end for**
- 10: **return** the solution with the lowest objective (2)

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*Grouped weights (proposed).* We posit that units fall into a small number  $G$  of latent segments whose members are well served by the same combination. The GDMA estimator solves

$$(\widehat{\gamma}, \widehat{\mathbf{W}}) = \arg \min_{\gamma \in \{1, \dots, G\}^N, \mathbf{W} = (\mathbf{w}_1, \dots, \mathbf{w}_G) \in \Delta_M^G} \sum_{i=1}^N \sum_{t=1}^T C_i(\mathbf{q}_{it}^\top \mathbf{w}_{\gamma_i}, y_{it}), \quad (2)$$

and unit  $i$  commits  $\mathbf{q}_{i,T+1}^\top \widehat{\mathbf{w}}_{\widehat{\gamma}_i}$ .  $G = 1$  gives pooled weights and  $G = N$  gives unit-specific weights, so the number of segments indexes the whole range between the two extremes. Three features separate (2) from existing proposals. First, the criterion is the decision cost in the planner’s own units (money or energy), so units are grouped by *how they should act*, not by how their forecasts look. Second, memberships and weights are estimated jointly; a two-step procedure that clusters noisy unit-specific weights and then re-estimates is a natural but inferior alternative (Section 5). Third, because each unit keeps its own critical ratio and its own candidate margins, units in one segment share a combination rule but not a commitment level.

### 3.3. Computation

Problem (2) is a mixed-integer problem, but it has the structure of  $k$ -means with a non-Euclidean, data-dependent distortion, as in the grouped fixed-effects estimator of Bonhomme and Manresa (2015). Given memberships, the weights of each segment solve a pooled convex problem on the simplex; given weights, each unit joins the segment whose weights give it the lowest in-window cost. Algorithm 1 alternates the two steps. The weight step is a linear program; for speed we minimise a softplus-smoothed check function,  $\rho_\tau^\varepsilon(r) = \tau r + \varepsilon \log(1 + e^{-r/\varepsilon})$ , with bandwidth  $\varepsilon$  equal to 2% of the unit’s typical decision error, by sequential quadratic programming. On our data the smoothed solution is within 0.01% of the exact linear-programming optimum in cost. Each iteration weakly decreases the objective, so the algorithm converges to a local solution in finitely many steps; we run several starts and keep the best. Starting values are drawn by a  $k$ -means++ rule (Arthur and Vassilvitskii, 2007) in *regret space*: the next seed unit is drawn with probability proportional to the extra cost that units would incur if forced to use the closest existing seed’s own weights. In rolling applications the memberships of the previous window provide an additional warm start.

### 3.4. Soft segments

Segments need not be exact. When units within a segment are similar but not identical, a unit may benefit from a rule that sits between its own estimated weights and those of its segment. *Soft GDMA* uses

$$\widehat{\mathbf{w}}_i^{\text{soft}} = (1 - \kappa) \widehat{\mathbf{w}}_i + \kappa \widehat{\mathbf{w}}_{\gamma_i}, \quad \kappa \in [0, 1], \quad (3)$$

where  $\widehat{\mathbf{w}}_i$  are the unit-specific weights and  $\widehat{\mathbf{w}}_{\gamma_i}$  the GDMA weights of the unit's segment. Because the simplex is convex,  $\widehat{\mathbf{w}}_i^{\text{soft}} \in \Delta_M$ . The family nests every benchmark:  $\kappa = 1$  is GDMA,  $\kappa = 0$  is unit-specific weights,  $G = 1$  is shrinkage towards pooled weights (a decision-focused analogue of [Diebold and Shin, 2019](#)), and  $G = 1, \kappa = 1$  is pooled weights. The number of segments is chosen exactly as for GDMA (next subsection); given  $G$ ,  $\kappa$  is chosen on the same hold-out block from  $\{0, 0.25, 0.5, 0.75, 1\}$ . Choosing  $G$  first keeps the search small, which matters because hold-out costs are noisy when windows are short. Segment centres play the role of data-driven prior means, in the spirit of empirical Bayes shrinkage towards group-specific targets.

### 3.5. Choosing the number of segments

We split the estimation window into an early block (the first two thirds) and a late block. For each  $G \in \{1, \dots, G_{\max}\}$  we estimate (2) on the early block and record every unit's cost on the late block under its estimated membership and weights. Let  $G^\dagger$  minimise the total late-block cost. Following the one-standard-error rule ([Hastie et al., 2009](#)), we choose the smallest  $G$  whose total late-block cost exceeds that of  $G^\dagger$  by no more than one standard error of the summed unit-level cost differences, and then re-estimate on the full window. The rule respects time order and prefers parsimonious segmentations, which are easier to explain to the operators who would have to adopt them.

### 3.6. From commitment rules to storage siting

The combination rule is an operational decision, re-estimated every week. A planner also faces a slower, capital-intensive decision: where to install new flexible capacity. We link the two with a two-stage design. In the operational stage, the learned rule leaves each unit with hourly shortfalls  $s_{it} = (y_{it} - q_{it})^+$  and surpluses  $x_{it} = (q_{it} - y_{it})^+$ . In the planning stage, a battery with power  $P_i$  (MW) and energy  $\bar{h}P_i$  (MWh) installed in unit  $i$  discharges to cover shortfalls and recharges from committed-but-unused capacity, which has already been paid for. With state of charge  $e_{it}$  and round-trip efficiency  $\eta$ , the chronological dispatch is

$$d_{it} = \min\{P_i, s_{it}, e_{it}\}, \quad c_{it} = \min\{P_i, x_{it}, (\bar{h}P_i - e_{it} + d_{it})/\eta\}, \\ e_{i,t+1} = e_{it} - d_{it} + \eta c_{it},$$

and every MWh discharged avoids the underage cost  $u_i$ . The annual value of battery power,  $V_i(P)$ , equal to  $u_i$  times the expected energy discharged per year, is non-decreasing and typically concave in  $P$ , because each additional MW of power is called on only in rarer and larger shortfalls; where the estimated curve is not concave on the grid we replace later increments by the running minimum of the marginal values, which yields a concave lower bound. Given an annualised cost  $\kappa$  per MW of battery power, the planner solves

$$\max_{P_1, \dots, P_N \geq 0} \sum_{i=1}^N \{V_i(P_i) - \kappa P_i\} \quad \text{subject to} \quad \sum_{i=1}^N P_i \leq B, \quad (4)$$

where  $B$  is a storage budget or deployment target (for example a national or continental procurement programme). Because each (concavified)  $V_i$  is concave, a greedy algorithm that adds power in small increments to the unit with the highest marginal value is optimal for (4); without the budget constraint, each unit simply installs power until its marginal value falls to  $\kappa$ . We estimate  $V_i$  on a grid of battery sizes from the out-of-sample shortfall and surplus paths of each commitment rule.

The two stages interact in two ways that matter for management. First, better commitment rules shrink the shortfalls that storage must cover, so part of the storage requirement can be met by better forecast combination instead of new hardware; we measure this *storage equivalent* of better forecasting as the battery power that the status-quo rule needs to reach the cost of the better rule without storage. Second, the siting decision depends on the rule: batteries placed where the status-quo rule is weakest may be placed where they add little once operations improve.

### 3.7. Fairness across decision units

At continental scale the objective in (2) has a distributional consequence. Because costs are measured in money or energy, the largest systems dominate the criterion, and a segment rule that is excellent for a large system may be mediocre for a small or data-poor one that happens to share its segment. The same holds for storage: problem (4) places batteries where the value is highest in absolute terms. Energy-justice scholarship stresses that the distribution of the benefits of energy infrastructure matters in its own right (Sovacool and Dworkin, 2015), and the operations literature measures the efficiency cost of fair allocations by the *price of fairness* (Bertsimas et al., 2011). We therefore add two fair variants and audit every method for how its benefits are distributed.

*Fair soft GDMA (one area, one vote).* Let  $c_i^0$  be the in-window cost of unit  $i$  under the reference forecaster (candidate 1, the official forecast where one exists). Fair soft GDMA solves the soft GDMA problem with every unit’s costs divided by  $c_i^0$  (normalised to mean one), that is, with  $u_i$  and  $o_i$  replaced by  $\omega_i u_i$  and  $\omega_i o_i$ ,  $\omega_i \propto 1/c_i^0$ . Each unit then counts in proportion to its *relative* improvement, so that a 10% saving in a 70 MW municipal utility weighs as much as a 10% saving in a 90 GW regional market. This is the relative-cost analogue of group-robust learning (Sagawa et al., 2020); it changes only the weights of the criterion, not the algorithm.

*Fair storage siting.* We consider two fair versions of (4) over continents  $c$  (any grouping, such as countries or income groups, works the same way). *Proportional floors* require every continent to receive at least a share  $\lambda$  of its share of demand in the budget; the floors are filled greedily within each continent and the rest of the budget greedily across all units. *Maximin siting* allocates each increment of battery power to the continent whose relative benefit—the value captured divided by its commitment cost without storage—is currently the lowest, and within it to the unit with the highest marginal value, which implements a max–min notion of fairness (Bertsimas et al., 2011). The price of fairness is the share of the efficient programme’s value that each fair version gives up.

*Audit.* For every method we report the distribution across units of the ratio of its cost to the unit’s cost under the reference forecaster, the share of units made worse off than under the reference forecaster and than under their own best single candidate (chosen on the estimation window), the median ratio of the worst-served continent, and separate ratios for data-rich and data-poor systems.

## 4. Theoretical properties

This section studies the global minimiser of (2) for a given  $G$ . We show that (i) its expected decision cost is close to that of the best grouped policy, with an estimation error that separates the price of learning memberships from the price of learning weights; (ii) it is asymptotically optimal and, under a latent segment structure, attains the risk of the infeasible unit-specific oracle; and (iii) it recovers the segments with probability approaching one, in which case its weights coincide with those of an estimator that is told the segments. Proofs are in Appendix A. The scope of these results should be stated plainly. They concern the exact global minimiser of the grouped criterion for a fixed number of segments, under stationarity, bounded losses and a high-level concentration condition; the estimator used in the applications is computed by a local search, with  $G$  and the shrinkage intensity chosen on a hold-out, and the data are neither stationary nor independent across units. The theory therefore explains why grouping can beat both unit-specific and pooled weights and what it costs to learn the segments; it does not deliver numerical guarantees for the empirical estimator.

### 4.1. Setting and assumptions

Let  $\mathbf{z}_{it} = (y_{it}, \mathbf{q}_{it})$ . Define the population risk of unit  $i$ ,  $R_i(\mathbf{w}) = \mathbb{E} C_i(\mathbf{q}_{it}^\top \mathbf{w}, y_{it})$ , and for a grouped policy  $(\gamma, \mathbf{W})$  the average risk and its empirical analogue

$$R(\gamma, \mathbf{W}) = \frac{1}{N} \sum_{i=1}^N R_i(\mathbf{w}_{\gamma_i}), \quad \hat{R}(\gamma, \mathbf{W}) = \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T C_i(\mathbf{q}_{it}^\top \mathbf{w}_{\gamma_i}, y_{it}).$$

Let  $\mathcal{P}_G = \{1, \dots, G\}^N \times \Delta_M^G$  and  $R_G^* = \inf_{\mathcal{P}_G} R$ .

**Assumption 1** (Stationarity and boundedness). *For each  $i$ ,  $\{\mathbf{z}_{it}\}_t$  is strictly stationary,  $|y_{it}| \leq B$  and  $\|\mathbf{q}_{it}\|_\infty \leq B$  almost surely, and  $\max_i(u_i + o_i) \leq \bar{a}$ .*

Stationarity refers to the candidate commitments as well as demand; it holds, for instance, when every candidate is computed from a rolling window of fixed length of a stationary process. Boundedness is natural once demand is normalised by a unit's scale. Under Assumption 1 each loss lies in  $[0, L]$  with  $L = 2\bar{a}B$  and is Lipschitz in the weights:  $|C_i(\mathbf{q}^\top \mathbf{w}, y) - C_i(\mathbf{q}^\top \mathbf{w}', y)| \leq \bar{a}B \|\mathbf{w} - \mathbf{w}'\|_1$ .

**Assumption 2** (Concentration). *There are constants  $c > 0$  and an effective sample length  $T_e \leq T$  such that, for every fixed  $(\gamma, \mathbf{W}) \in \mathcal{P}_G$  and every  $x > 0$ ,*

$$\Pr(|\hat{R}(\gamma, \mathbf{W}) - R(\gamma, \mathbf{W})| > x) \leq 2 \exp(-c N T_e x^2 / L^2),$$

*and, for every  $i$  and fixed  $\mathbf{w}$ ,  $\Pr(|\hat{R}_i(\mathbf{w}) - R_i(\mathbf{w})| > x) \leq 2 \exp(-c T_e x^2 / L^2)$ .*

Assumption 2 is a high-level condition. It holds with  $T_e = T$  when observations are independent (Hoeffding's inequality) and, when units are independent and each  $\{\mathbf{z}_{it}\}_t$  is geometrically  $\beta$ -mixing, with  $T_e = T/(C \log T)$  by the blocking argument of Yu (1994) or the Bernstein inequality of Merlevède et al. (2011). Cross-sectional dependence through common shocks (weather, in our application) reduces the effective number of independent units; the results continue to hold with  $N$  replaced by an effective cross-sectional size.

#### 4.2. An oracle inequality

**Theorem 1** (Oracle inequality). *Let Assumptions 1–2 hold and let  $(\hat{\gamma}, \widehat{\mathbf{W}})$  be a global minimiser of (2). For any  $\delta \in (0, 1)$ , with probability at least  $1 - \delta$ ,*

$$R(\hat{\gamma}, \widehat{\mathbf{W}}) \leq R_G^* + 2L \sqrt{\frac{GM \log(3NT_e) + N \log G + \log(2/\delta)}{c NT_e}} + \frac{4\bar{a}B}{NT_e}. \quad (5)$$

The square-root term has a transparent decomposition. Per unit, learning the segment membership costs  $\log G/T_e$ , while learning the segment weights costs  $GM \log(NT_e)/(NT_e)$ . The same argument applied to unit-specific weights ( $G = N$ ) gives  $M \log(NT_e)/T_e$ , and applied to pooled weights ( $G = 1$ ) gives  $M \log(NT_e)/(NT_e)$  but a larger approximation error  $R_1^* - R_N^*$ . Grouping therefore replaces the  $M$  per-unit parameters of unit-specific weights by a single discrete label that costs  $\log G$  rather than  $M \log(NT_e)$  to learn. This is the formal sense in which GDMA resolves the tension between the noise of unit-specific weights and the bias of pooled weights: with  $M = 7$  candidates,  $N = 46$  units,  $G = 4$  segments and an effective length of  $T_e = 28$  (one observation per day of a four-week window), the variance proxy under the square root is 0.23 for GDMA against 2.07 for unit-specific weights, a factor of three in the square-root term, while the approximation error of GDMA is zero whenever the segment structure is correct. With unknown constants  $L$  and  $c$  both bounds are loose at these sizes, so the comparison concerns how the two error terms scale, not the size of the bounds.

#### 4.3. Asymptotic optimality

Following the model-averaging literature (Hansen, 2007; Hansen and Racine, 2012; Zhang and Liu, 2023), call a weight choice asymptotically optimal if the ratio of its risk to the lowest attainable risk in its class tends to one.

**Corollary 1** (Asymptotic optimality). *Let the assumptions of Theorem 1 hold, with  $G$  and  $M$  possibly growing, and suppose  $R_G^* \geq \kappa > 0$ . If  $GM \log(NT_e)/(NT_e) \rightarrow 0$  and  $\log G/T_e \rightarrow 0$ , then  $R(\hat{\gamma}, \widehat{\mathbf{W}})/R_G^* \rightarrow_p 1$ . If, in addition, there are memberships  $\gamma^0$  with  $G_0 \leq G$  segments such that  $\arg \min_{\mathbf{w}} R_i(\mathbf{w})$  depends on  $i$  only through  $\gamma_i^0$ , then  $R_G^* = R_N^* = N^{-1} \sum_i \min_{\mathbf{w}} R_i(\mathbf{w})$  and*

$$\frac{R(\hat{\gamma}, \widehat{\mathbf{W}})}{N^{-1} \sum_i \min_{\mathbf{w} \in \Delta_M} R_i(\mathbf{w})} \rightarrow_p 1.$$

The second statement is the relevant one for practice: under a segment structure, GDMA attains the risk of the infeasible benchmark in which every unit knows its own optimal weights, although it estimates only  $G(M-1)$  continuous parameters. Unit-specific weights attain the same limit only if  $M \log(NT_e)/T_e \rightarrow 0$ , a much stronger requirement when windows are short because the system is non-stationary and old data are of limited use. The condition  $R_G^* \geq \kappa$  excludes the degenerate case of perfect foresight, which never occurs in commitment problems.

#### 4.4. Recovery of the segments

**Assumption 3** (Segment structure). *(a) There are  $\gamma^0 \in \{1, \dots, G\}^N$  and distinct  $\mathbf{w}_1^0, \dots, \mathbf{w}_G^0 \in \Delta_M$  such that, for all  $i$  and all  $\mathbf{w} \in \Delta_M$ ,  $R_i(\mathbf{w}) - R_i(\mathbf{w}_{\gamma_i^0}^0) \geq \lambda \|\mathbf{w} - \mathbf{w}_{\gamma_i^0}^0\|_2^2$  for some  $\lambda > 0$ . (b)  $\min_{g \neq h} \|\mathbf{w}_g^0 - \mathbf{w}_h^0\|_2 \geq d > 0$  and every segment contains at least  $\pi N$  units,  $\pi > 0$ .*

Part (a) is a quadratic-growth condition. It holds when the conditional density of demand given the candidates is bounded away from zero near the optimal commitment and the candidates are not collinear, because the Hessian of  $R_i$  is then  $(u_i + o_i) \mathbb{E}[f_{y|q}(\mathbf{q}^\top \mathbf{w}) \mathbf{q} \mathbf{q}^\top]$ , which is positive definite on the tangent space of the simplex. Part (b) requires segments to be genuinely different and non-negligible, as in [Bonhomme and Manresa \(2015\)](#).

**Theorem 2** (Segment recovery and oracle equivalence). *Let Assumptions 1–3 hold with  $G$  equal to the true number of segments, and let  $N, T_e \rightarrow \infty$  with  $\log N/T_e \rightarrow 0$ . Then, up to a relabelling of segments, (i)  $\max_g \|\widehat{\mathbf{w}}_g - \mathbf{w}_g^0\|_2 \rightarrow_p 0$ ; (ii)  $\Pr(\widehat{\gamma} \neq \gamma^0) \leq N C_M \exp(-c' T_e) + o(1) \rightarrow 0$  for constants  $C_M, c' > 0$  that do not depend on  $N$  or  $T$ ; (iii) with probability approaching one,  $\widehat{\mathbf{W}}$  equals the infeasible estimator  $\widetilde{\mathbf{W}} = \arg \min_{\mathbf{W}} \widehat{R}(\gamma^0, \mathbf{W})$  that knows the segments.*

Part (iii) is an exact equivalence, not an asymptotic approximation: on the event that memberships are recovered, the joint minimiser and the known-segment minimiser are the same object. Any distributional result available for the known-segment estimator therefore transfers to GDMA without modification. The recovery condition  $\log N/T_e \rightarrow 0$  is mild: the number of units may grow polynomially in the window length.

*Soft GDMA..* For fixed  $(G, \kappa)$ ,  $R_i$  is convex in  $\mathbf{w}$ , so the risk of (3) satisfies  $R_i(\widehat{\mathbf{w}}_i^{\text{soft}}) \leq (1 - \kappa)R_i(\widehat{\mathbf{w}}_i) + \kappa R_i(\widehat{\mathbf{w}}_{\widehat{\gamma}_i})$ . Averaging over units, the excess risk of soft GDMA is at most the  $\kappa$ -weighted average of the excess risks of unit-specific weights and of GDMA, each controlled by Theorem 1. Under a segment structure, therefore, any  $\kappa$  bounded away from zero inherits the faster rate of GDMA for the  $\kappa$  share of the combination, while  $\kappa < 1$  buys robustness when segments are only approximate. These statements hold for fixed  $(G, \kappa)$ . We do not analyse the data-dependent choice of  $(G, \kappa)$  on the hold-out, nor why soft GDMA with an estimated  $\kappa$  can outperform both of its limits; a formal treatment of the hold-out rule under serial and cross-sectional dependence is left for future work.

*Weighted and fair criteria..* All results of this section hold for criteria in which unit  $i$ 's costs are multiplied by a weight  $\omega_i \in [\underline{\omega}, \bar{\omega}]$  with  $0 < \underline{\omega} \leq \bar{\omega} < \infty$ : the weights only rescale  $u_i$  and  $o_i$ , so Assumptions 1–3 hold with  $\bar{a}$  replaced by  $\bar{\omega} \bar{a}$  and  $\lambda$  by  $\underline{\omega} \lambda$ . Fair soft GDMA uses data-dependent weights  $\omega_i \propto 1/c_i^0$ ; treating them as fixed within a window, Theorem 1 bounds its weighted excess risk. Under the segment structure of Corollary 1, the weighted and unweighted criteria have the same minimiser, because every unit attains its own optimal weights, and since each unit's excess risk is non-negative the unweighted excess risk is at most  $1/\underline{\omega}$  times the weighted one (weights normalised to mean one). Fairness then costs nothing asymptotically, although the factor  $1/\underline{\omega}$  can be large when the areas' costs differ by orders of magnitude. When segments are only approximate, the fair criterion trades some total cost for a more even distribution of the gains; this is the price of fairness that we measure empirically.

## 5. Monte Carlo evidence

### 5.1. Design

The simulation examines the finite-sample behaviour of the estimator as it is used in the applications, with  $G$  and  $\kappa$  chosen on a hold-out; it does not verify the rates of Section 4, which concern the exact minimiser for a fixed  $G$ . It is deliberately simpler than the applications: units are independent, the data are stationary, and costs are linear, so

Table 2: Monte Carlo results: out-of-sample newsvendor cost relative to the unit-specific oracle (means over 50 replications), average selected number of segments  $\bar{G}$  of GDMA, and adjusted Rand index (ARI; [Hubert and Arabie, 1985](#)) between estimated and true segments for GDMA and two-step clustering. Bold: lowest in the row.

| Design      | $N$ | $T$ | Out-of-sample cost / oracle |              |              |              |        |       |              | $\bar{G}$ | ARI  |        |
|-------------|-----|-----|-----------------------------|--------------|--------------|--------------|--------|-------|--------------|-----------|------|--------|
|             |     |     | EW                          | Pooled       | Per-unit     | Shrink       | 2-step | GDMA  | Soft         |           | GDMA | 2-step |
| Grouped     | 50  | 48  | 1.496                       | 1.414        | 1.084        | 1.090        | 1.055  | 1.050 | <b>1.041</b> | 3.4       | 0.87 | 0.75   |
|             | 50  | 168 | 1.491                       | 1.410        | 1.033        | 1.033        | 1.031  | 1.030 | <b>1.023</b> | 3.7       | 0.88 | 0.79   |
|             | 50  | 672 | 1.481                       | 1.398        | 1.013        | 1.013        | 1.019  | 1.018 | <b>1.011</b> | 5.3       | 0.68 | 0.67   |
|             | 200 | 48  | 1.491                       | 1.418        | 1.086        | 1.089        | 1.049  | 1.040 | <b>1.035</b> | 3.1       | 0.93 | 0.79   |
|             | 200 | 168 | 1.492                       | 1.419        | 1.032        | 1.032        | 1.028  | 1.023 | <b>1.019</b> | 5.2       | 0.71 | 0.73   |
|             | 200 | 672 | 1.491                       | 1.416        | 1.012        | 1.012        | 1.021  | 1.017 | <b>1.011</b> | 5.8       | 0.62 | 0.66   |
| Homogeneous | 50  | 48  | 1.321                       | <b>1.027</b> | 1.097        | <b>1.027</b> | 1.032  | 1.028 | 1.028        | 1.1       | —    | —      |
|             | 50  | 168 | 1.322                       | 1.025        | 1.036        | <b>1.020</b> | 1.026  | 1.024 | <b>1.021</b> | 1.3       | —    | —      |
|             | 50  | 672 | 1.322                       | 1.025        | 1.015        | 1.013        | 1.017  | 1.016 | <b>1.012</b> | 2.6       | —    | —      |
|             | 200 | 48  | 1.321                       | 1.026        | 1.094        | <b>1.025</b> | 1.027  | 1.026 | <b>1.025</b> | 1.0       | —    | —      |
|             | 200 | 168 | 1.320                       | 1.025        | 1.038        | <b>1.020</b> | 1.026  | 1.024 | <b>1.020</b> | 1.3       | —    | —      |
|             | 200 | 672 | 1.321                       | 1.025        | 1.014        | 1.013        | 1.017  | 1.015 | <b>1.012</b> | 2.9       | —    | —      |
| Continuous  | 50  | 48  | 1.206                       | 1.148        | 1.091        | 1.075        | 1.090  | 1.091 | <b>1.068</b> | 2.7       | —    | —      |
|             | 50  | 168 | 1.206                       | 1.148        | 1.035        | 1.036        | 1.051  | 1.055 | <b>1.031</b> | 4.0       | —    | —      |
|             | 50  | 672 | 1.203                       | 1.145        | <b>1.014</b> | <b>1.014</b> | 1.035  | 1.035 | <b>1.014</b> | 5.4       | —    | —      |
|             | 200 | 48  | 1.202                       | 1.146        | 1.094        | 1.075        | 1.090  | 1.082 | <b>1.062</b> | 3.1       | —    | —      |
|             | 200 | 168 | 1.205                       | 1.149        | 1.035        | 1.038        | 1.050  | 1.049 | <b>1.030</b> | 4.9       | —    | —      |
|             | 200 | 672 | 1.206                       | 1.150        | <b>1.013</b> | <b>1.013</b> | 1.040  | 1.037 | <b>1.014</b> | 5.9       | —    | —      |

it isolates the trade-off between the noise of unit-specific weights and the bias of pooled weights.

We simulate  $N \in \{50, 200\}$  units and  $M = 5$  candidate commitment rules. Unit  $i$  has demand  $y_{it} = \mu_i + x_{it}$ , with  $\mu_i \sim U[20, 100]$  and  $x_{it}$  a stationary AR(1) process with coefficient 0.5 and standard deviation 5, and critical ratio  $\tau_i \sim U[0.6, 0.95]$  ( $u_i = \tau_i / (1 - \tau_i)$ ,  $o_i = 1$ ). Candidate  $m$  forecasts  $f_{itm} = y_{it} - e_{itm}$ , where the errors follow AR(1) processes with coefficient 0.5 and innovations with unit-specific scales  $s_{im}$  and skewness signs  $k_{im} \in \{-1, 0, +1\}$  (a centred gamma distribution for  $\pm 1$ , a normal distribution for 0). Each candidate becomes a commitment by adding the  $\tau_i$ -quantile of its errors in a pre-sample of 200 periods. Three designs define the heterogeneity: *grouped*—every unit belongs to one of three segments with segment-specific error profiles  $(s_i, k_i)$ , so that different candidates are best in different segments; *homogeneous*—all units share the first profile; *continuous*—each unit’s error scales are a Dirichlet(1,1,1) mixture of the three profiles, so that no two units share their optimal weights and GDMA’s segment structure is misspecified. Weights are estimated on  $T \in \{48, 168, 672\}$  periods and evaluated on 1,000 new periods. We report the out-of-sample cost relative to an infeasible oracle that uses, for every unit, the exact cost-minimising weights on the evaluation sample itself. Each cell uses 50 replications.

## 5.2. Results

Table 2 supports the theory and clarifies when grouping pays.

*Soft GDMA is the most reliable method.* It has the lowest cost, or is within 0.1 percentage points of the lowest, in every one of the 18 cells. Its advantage is largest when windows are short: with true segments and  $T = 48$  it is 3.5–4.1% above the oracle

against 8.4–8.6% for unit-specific weights, and even under continuous heterogeneity, where segments are misspecified, it beats unit-specific weights (6.2–6.8% against 9.1–9.4% above the oracle) and shrinkage towards pooled weights (7.5%). When units share one error profile it matches pooled weights for short windows and beats them for long ones, where the heterogeneity in critical ratios makes some unit-level adjustment worthwhile.

*Hard segments trade bias for variance exactly as Theorem 1 predicts.* Plain GDMA beats unit-specific weights in 96–100% of the replications with true segments and  $T = 48$ , and in all replications with homogeneous units for  $T \leq 168$ . With long windows ( $T = 672$ ), unit-specific weights are estimated precisely and become slightly better (1.2–1.3% against 1.7–1.8% above the oracle). Here GDMA selects five to six segments rather than three, because units within a true segment have different critical ratios and therefore different optimal weights: with enough data the hold-out rule correctly detects this finer structure, which is also why the ARI falls from about 0.9 at  $T = 48$  to about 0.65 at  $T = 672$ . Soft GDMA removes this residual bias through unit-level shrinkage and is never worse than unit-specific weights by more than 0.1 percentage points.

*Joint estimation recovers segments better than clustering estimated weights.* With short windows, the ARI of GDMA is 0.87–0.93, against 0.75–0.79 for two-step  $k$ -means on noisy unit-level weights, and GDMA’s cost is lower in every grouped cell with  $T \leq 168$ . When units share one error profile, the one-standard-error rule selects a single segment in most replications with  $T \leq 168$  ( $\bar{G} = 1.0$ –1.3).

## 6. Day-ahead capacity commitment on the U.S. grid

### 6.1. Data

We use the complete public record of the U.S. Energy Information Administration’s Hourly Electric Grid Monitor (Form EIA-930; [U.S. Energy Information Administration, 2026](#)), downloaded on 29 September 2026. For every balancing authority (BA) in the lower 48 states the file reports hourly demand, the BA’s own day-ahead demand forecast and total interchange with neighbouring BAs. We use the period 1 January 2019 to 26 September 2026 in UTC hours and exclude the fourteen regional aggregates. Following [Ruggles et al. \(2020\)](#), non-positive values and values outside  $[0.3, 3]$  times the median of the preceding seven days are treated as missing, and gaps of up to three hours are interpolated; the filter is one-sided, so it uses no future data, and the official forecast is filtered against its own history rather than against realised demand. We keep BAs with at least 97% valid demand, at least 93% valid official forecasts, mean demand above 50 MW and a median absolute official forecast error below 15% of median demand. Forty-six BAs remain (Table C.20 in Appendix C); they range from PJM (mean demand 92.5 GW) to the City of Homestead (71 MW), include all seven independent system operators, and account for 97% of lower-48 demand. Ten BAs are dropped, mostly small systems with long gaps or implausible official forecasts; 0.27% of the retained BA-hours are missing.

### 6.2. Decision problem and cost calibration

On the morning of day  $d - 1$ , before the day-ahead gate closure, BA  $i$  commits hourly capacity  $q_{i,d,h}$ ,  $h = 0, \dots, 23$ , for the (UTC) operating day  $d$ . At that time demand is observed only up to the end of day  $d - 2$ : the UTC day  $d - 1$  ends in the afternoon or evening of local day  $d - 1$  in the United States, after any day-ahead deadline. The information set therefore contains realised demand up to the end of day  $d - 2$ , the realised costs of every rule up to the same day, and the BA’s published day-ahead forecast for

day  $d$ ; every candidate, margin and estimation window below respects this timing. This is a conservative convention: an operator could also use the hours of day  $d - 1$  observed before the gate closure, which would favour the data-driven candidates slightly. Excess commitment costs  $o_i = 1$  per MWh, which normalises all costs to units of the per-MWh cost of idle committed capacity. Shortfalls cost  $u_i = \tau_i / (1 - \tau_i)$ , where the critical ratio  $\tau_i$  is meant to reflect how easily the BA can lean on its neighbours. As a transparent proxy we use the variability of interchange in the pre-sample year 2019—the range between the 5th and 95th percentiles of hourly total interchange, divided by mean demand—and let  $\tau_i$  decline linearly in its rank from 0.95 for the BA with the least variable interchange to 0.80 for the one with the most. Interchange variability is not transfer capability, and shortfall costs in practice depend on market prices and system conditions; the proxy only orders BAs plausibly, and Section 6.7 reports the results under common critical ratios from 0.5 to 0.95 (and Section 7 for the global panel under a scarcity-level ratio of 0.99). ERCOT, which is almost islanded, receives  $\tau = 0.95$  ( $u_i = 19$ ); the Chelan County PUD in the densely interconnected Pacific Northwest receives  $\tau = 0.80$  ( $u_i = 4$ ).

### 6.3. Candidate rules

Seven candidate forecasts are produced for every BA, day and hour, each using only the information available at the decision time: (1) the BA’s official day-ahead forecast; (2) persistence (same hour of  $d - 2$ ); (3) weekly persistence (same hour of  $d - 7$ ); (4) the mean of the same hour over  $d - 2, \dots, d - 8$ ; (5) the mean of the same hour on  $d - 7, d - 14$  and  $d - 21$ ; (6) a BA-and-hour-specific linear regression on the  $d - 2$  and  $d - 7$  values, the official forecast and weekday dummies, estimated on a rolling 56-day window and refitted weekly; and (7) a single global LightGBM model (Ke et al., 2017) across all BAs and hours, trained on lags, calendar variables and the official forecast, all scaled by the BA’s recent mean demand, refitted every 28 days on the preceding 365 days. Every training window ends at day  $d_0 - 2$  for forecasts made at origin  $d_0$ . Model (7) is the kind of machine-learning forecaster a central planner could run for every BA at low cost. Each forecast  $f^{(m)}$  becomes a candidate commitment by adding its own safety margin,

$$q_{i,d,h}^{(m)} = f_{i,d,h}^{(m)} \{1 + \hat{Q}_{\tau_i}(e_i^{(m)})\},$$

where  $\hat{Q}_{\tau_i}(e_i^{(m)})$  is the empirical  $\tau_i$ -quantile of the relative errors  $(y - f^{(m)}) / \max\{f^{(m)}, 0.2\bar{y}_{i,t}\}$  of forecast  $m$  for BA  $i$  over the 28 days ending at  $d - 2$ ; the floor at 20% of the BA’s mean demand  $\bar{y}_{i,t}$  over the week before the error’s day  $t$  prevents an occasional implausibly small forecast from producing an unbounded margin. The candidate rules are available from February 2020, so the evaluation period runs from 1 April 2020 to 26 September 2026: 2,342 days, 335 weekly re-estimations and 2.58 billion MWh of demand.

### 6.4. Competing methods

Every seven days, at origin  $d_0$ , each method re-estimates its combination from the  $W$  most recent days whose outcomes are known,  $d_0 - W - 1, \dots, d_0 - 2$ ,  $W \in \{7, 14, 28, 56\}$  (each day contributes 24 hourly observations per BA), and commits for the following seven days. Besides GDMA and soft GDMA ( $G_{\max} = 6$ ) we consider: the official-forecast policy (candidate 1 alone), the reference rule in the U.S. analysis—a stylised version of committing on the operator’s own forecast, since actual commitment runs through market clearing, reliability unit commitment and reserve procurement; equal weights over

Table 3: Out-of-sample system-wide commitment cost, 1 April 2020 to 26 September 2026, relative to committing on the official day-ahead forecasts.  $W$  is the length of the estimation window. Bold: lowest cost in the column. Stars: Diebold–Mariano test of equal daily system cost against soft GDMA with Newey–West variance (7 lags); \*, \*\*, \*\*\*:  $p < 0.10, 0.05, 0.01$ .

| Method                        | System cost relative to the official-forecast policy |                  |                  |                  | Share of BAs improved <sup>a</sup> |
|-------------------------------|------------------------------------------------------|------------------|------------------|------------------|------------------------------------|
|                               | $W = 7$<br>days                                      | $W = 14$<br>days | $W = 28$<br>days | $W = 56$<br>days |                                    |
| Official forecast             | 1.0000***                                            | 1.0000***        | 1.0000***        | 1.0000***        | 0.00                               |
| Equal weights                 | 1.6901***                                            | 1.6902***        | 1.6902***        | 1.6902***        | 0.09                               |
| Best single (per BA)          | 0.8942***                                            | 0.8692***        | 0.8585***        | 0.8499***        | 0.50                               |
| Pooled DF weights             | <b>0.8556***</b>                                     | 0.8492           | 0.8446***        | 0.8400***        | 0.52                               |
| Per-BA DF weights             | 0.8898***                                            | 0.8581***        | 0.8422***        | 0.8308***        | 0.54                               |
| Shrinkage to pooled           | 0.8661                                               | 0.8513**         | 0.8385***        | 0.8297**         | 0.54                               |
| Two-step $k$ -means           | 0.8589***                                            | <b>0.8446**</b>  | <b>0.8348†</b>   | <b>0.8286</b>    | 0.57                               |
| Forecast-then-commit, per BA  | 0.9300***                                            | 0.8929***        | 0.8673***        | 0.8480***        | 0.50                               |
| Forecast-then-commit, grouped | 0.8824***                                            | 0.8716***        | 0.8548***        | 0.8471***        | 0.52                               |
| GDMA                          | 0.8642**                                             | 0.8485           | 0.8356†          | 0.8300           | 0.59                               |
| Quantile-regression comb.     | 1.2970***                                            | 1.0186***        | 0.8834***        | 0.8455***        | 0.52                               |
| GDMA, minimum hold-out rule   | 0.8735***                                            | 0.8506           | 0.8383***        | 0.8300*          | 0.59                               |
| <b>Soft GDMA</b>              | 0.8673                                               | 0.8492           | 0.8355†          | 0.8288           | 0.63                               |

<sup>a</sup> Share of the 46 BAs whose own cost is lower than under the official-forecast policy ( $W = 28$ ).

the seven candidates; the best single candidate per BA over the window; pooled decision-focused weights; per-BA decision-focused weights (the decision-focused analogue of per-series forward validation); shrinkage of per-BA weights towards pooled weights with the shrinkage intensity chosen on the hold-out (a decision-focused version of [Diebold and Shin, 2019](#)); two-step clustering, which runs  $k$ -means on the per-BA weight vectors and re-estimates pooled weights within clusters, with the number of clusters chosen by the same hold-out and one-standard-error rule as GDMA; a linear quantile-regression combination, which regresses demand on the seven point forecasts at each BA’s critical ratio  $\tau_i$  without the simplex constraint (the natural decision-focused combination of [Wang et al., 2019](#)); and two *forecast-then-commit* methods, which estimate per-BA or grouped combination weights by minimising scaled squared forecast error and then add the  $\tau_i$ -quantile of the combined forecast’s relative errors over the same 28 days that set the candidates’ margins. The last two isolate the value of estimating the combination for the decision rather than for accuracy. All hold-out choices ( $G$ ,  $\kappa$  and the shrinkage intensity) use the same split of the estimation window: every third day, counted back from the end of the window, is a validation day, so that the validation days span the window and cycle through the days of the week across origins.

## 6.5. Results

Table 3 contains the main result. Four findings stand out.

*First, deciding for the decision pays.* Soft GDMA lowers the system-wide commitment cost by 13–17% relative to committing on the official forecasts, depending on the window, and every decision-focused combination on the simplex lowers it by 11–17%; soft GDMA’s reduction is significant at the 1% level for every window. The forecast-then-commit methods, which combine the same seven forecasts for accuracy and then add a quantile margin, save 7–15%, and they are significantly more expensive than soft GDMA for every window length; the gap is largest when windows are short ( $W = 7$ : 0.930 and

0.882 against 0.867), because accuracy-optimal weights are chosen without regard to the asymmetric costs. The unconstrained quantile-regression combination, the most direct decision-focused alternative, is competitive only with long windows (0.846 at  $W = 56$ ) and very poor with short ones (1.297 at  $W = 7$ ): with seven highly collinear forecasts and a few weeks of data, freeing the weights from the simplex costs more in variance than it gains in flexibility. Equal weights are the worst choice: they give weight to the naive calendar rules and raise cost by 69%.

*Second, neither extreme is right.* Per-BA weights are expensive when windows are short (0.890 at  $W = 7$ , 4.0% above pooled weights), pooled weights are expensive when windows are long (0.840 at  $W = 56$ , 1.4% above the best grouped rule), and the ranking of the two reverses between  $W = 14$  and  $W = 28$ . This is the bias–variance trade-off that motivates GDMA.

*Third, grouping helps, but the three grouped rules are statistically indistinguishable.* For  $W \geq 14$  the three cheapest methods are the grouped ones—two-step clustering, GDMA and soft GDMA—and at  $W = 28$  they form the 90% model confidence set of Hansen et al. (2011) (marked  $\dagger$  in Table 3), which excludes shrinkage towards pooled weights (MCS  $p$ -value 0.03), per-BA and pooled weights and all other rules. The differences among the three are below 0.1% at  $W = 28$  and  $W = 56$  and below 0.6% at  $W = 14$ ; two-step clustering is marginally the cheapest of them, and with a one-week window pooled weights are cheapest overall (0.856, against 0.859–0.867 for the grouped rules). Soft GDMA improves on the official forecast for the largest share of BAs (63%, against 57–59% for the other grouped rules and 54% for shrinkage), and choosing  $G$  by the one-standard-error rule rather than the plain hold-out minimum lowers the cost of GDMA for short windows (0.864 against 0.874 at  $W = 7$ ). The evidence therefore supports decision-focused combination with cross-sectional structure; which of the grouped variants a planner uses matters much less.

*Fourth, the savings are economically meaningful.* At  $W = 28$  the official-forecast policy costs 0.0779 units of idle-capacity cost per MWh of demand and soft GDMA 0.0651, a saving of 0.0128 units per MWh. Over the roughly 4.0 billion MWh of annual demand in our panel (25.8 billion MWh over the evaluation period), and valuing idle committed capacity at an illustrative \$10 per MWh, this is roughly \$500 million a year for the system as a whole under the stylised cost model, obtained by re-weighting forecasts that the operators already produce. The saving is not driven by a few operators, but it is not universal either: 63% of BAs are individually better off than under their own official forecast (last column of Table 3), and the official forecast remains the better rule for the rest.

Table 4 and Figure 1 confirm the picture. With every BA weighted equally rather than by its size, soft GDMA is the cheapest method, so the results are not an artefact of the three largest ISOs. The decision-focused simplex methods hit the target shortfall frequency closely (12.0–12.5% against a target of 12.5%, the average of  $1 - \tau_i$ ), whereas the official-forecast policy, the forecast-then-commit methods and the quantile-regression combination fall short too often (13.7–15.9%). The cumulative saving builds up steadily rather than in a few episodes.

## 6.6. Segments

GDMA uses a single segment in 16% of the weeks and two to four segments in two thirds of them (Figure 2, left); soft GDMA shrinks towards the segment rule with an average intensity of only  $\kappa = 0.14$ , so in the U.S. panel it mostly refines unit-level weights.

Table 4: Secondary metrics for  $W = 28$  days.

| Method                            | Scale-free cost <sup>b</sup><br>(rel. to EW) | Shortfall<br>frequency | DM statistic<br>vs soft GDMA <sup>c</sup> |
|-----------------------------------|----------------------------------------------|------------------------|-------------------------------------------|
| Official forecast                 | 0.6967                                       | 0.137                  | 23.68                                     |
| Equal weights                     | 1.0000                                       | 0.089                  | 36.71                                     |
| Best single (per BA)              | 0.6131                                       | 0.142                  | 7.46                                      |
| Pooled DF weights                 | 0.6122                                       | 0.120                  | 3.38                                      |
| Per-BA DF weights                 | 0.6030                                       | 0.121                  | 8.17                                      |
| Shrinkage to pooled               | 0.6008                                       | 0.120                  | 4.32                                      |
| Two-step $k$ -means               | 0.6035                                       | 0.124                  | -0.39                                     |
| Forecast-then-commit, per BA      | 0.6146                                       | 0.149                  | 5.47                                      |
| Forecast-then-commit, grouped     | 0.6115                                       | 0.143                  | 4.18                                      |
| GDMA                              | 0.6026                                       | 0.125                  | 0.11                                      |
| Quantile-regression comb.         | 0.6166                                       | 0.159                  | 7.84                                      |
| GDMA, minimum hold-out rule       | 0.6007                                       | 0.125                  | 2.95                                      |
| <b>Soft GDMA</b>                  | 0.5984                                       | 0.121                  | —                                         |
| Target (mean $\tau_i$ across BAs) |                                              | 0.125                  |                                           |

<sup>b</sup> Each BA’s cost divided by its mean hourly demand, averaged over BAs with equal weight, relative to equal weights. <sup>c</sup> Diebold–Mariano statistic; positive values: the method has higher daily system cost than soft GDMA.

The co-membership matrix (right) reveals two stable segments. The first contains 16 BAs, among them PJM, CAISO, ISO New England, Florida Power & Light, Arizona Public Service, Nevada Power, PacifiCorp East and the Southern Company; their combinations rely mostly on the learned models (weights of 0.38 on the hourly regression and 0.39 on LightGBM) and put 0.20 directly on the official forecast. The second contains 29 BAs, including MISO, ERCOT, NYISO, Southwest Power Pool, TVA, Bonneville and Duke Energy, and puts 0.44 directly on the official forecast; the Los Angeles Department of Water and Power often forms a segment of its own. The two main segments have similar average critical ratios (0.886 and 0.868), so the segmentation is not a relabelling of cost asymmetries; nor does it follow geography or market design, since both contain ISOs and vertically integrated utilities from the East and the West. What separates them is how much the operator’s own forecast adds to what can be learned from recent demand—information that a planner cannot read off the organisational chart of the grid, but that the data reveal.

### 6.7. Extreme events and robustness

Table 5 examines three extreme episodes. During Winter Storm Uri and the summer heat of 2023, the decision-focused combinations cost 4–7% and 16–19% less than the official-forecast policy. The Uri comparison should be read with care: ERCOT, and on two days SPP and MISO, shed load ([FERC, NERC and Regional Entity Staff, 2021](#)), and EIA-930 reports served demand, so a commitment that anticipated true demand is scored as over-commitment on those days; Section 7 repeats the event analysis without the load-shed BA-days. Winter Storm Elliott is different: over the six days of the storm the operators’ forecasts were much better than every candidate rule, and every combination is 17–30% more expensive than the official policy. This is consistent with the storm’s nature—an abrupt cold snap with large forced outages ([FERC, NERC and Regional Entity Staff, 2023](#))—and with the fact that weights learned from the previous weeks cannot

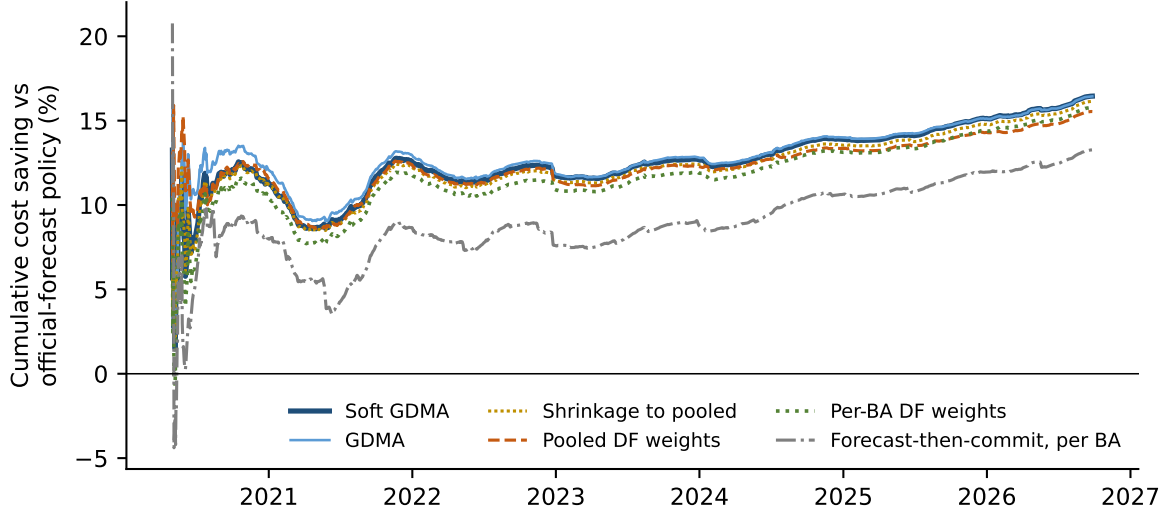


Figure 1: Cumulative system-wide cost saving relative to the official-forecast policy ( $W = 28$ ).

anticipate a regime change. Section 7 adds candidates driven by archived deterministic and ensemble weather forecasts and examines whether they narrow this gap.

Table 6 replaces the calibrated critical ratios by a common  $\tau$  for every BA, from the symmetric case  $\tau = 0.5$  to the highly asymmetric  $\tau = 0.95$ . The level of the savings changes little—soft GDMA saves 17–19% relative to the official-forecast policy—and neither does the ranking: two-step clustering is cheapest in every column, soft GDMA and GDMA follow within 0.5%, and shrinkage, per-BA and pooled weights come next. The advantage of decision-focused over forecast-then-commit combination grows with the asymmetry, from 1.4% at  $\tau = 0.5$  to 2.8% at  $\tau = 0.95$  relative to the better forecast-then-commit method, as expected: the more asymmetric the cost, the more the best combination for the decision departs from the most accurate one.

## 7. Capacity commitment on five continents

### 7.1. A global panel of grid areas

We assemble public hourly demand data for 125 grid areas on five continents (Table 7). *North America* is represented by the 46 U.S. balancing authorities of Section 6 (EIA Form 930, 2019–2026), the only systems for which the operators’ own day-ahead forecasts are part of the data; European transmission system operators and AEMO also publish day-ahead forecasts, but not through the sources we use, so outside the United States the reference rule is a model forecast (Section 7.2). *Europe* contributes 33 national systems from the Energy-Charts platform of Fraunhofer ISE ([Fraunhofer Institute for Solar Energy Systems ISE, 2026](#)), 2018–2026, including cross-border trading flows. *Oceania* contributes the five regions of Australia’s National Electricity Market ([Australian Energy Market Operator, 2026](#)), 2018–2026. *Asia* contributes the 31 provinces of mainland China for 2018 ([Wu and Kan, 2023](#)), the four grid areas of Taiwan for 2018 to mid-2022 ([Fu and Chen, 2023](#)) and the five regions of Thailand for 2023–2024 ([Bunnak, 2025](#)). *Africa* is represented by Algeria (Sonelgaz), 2018 to January 2020 ([Rania, 2023](#)). All series are converted to hourly UTC, cleaned as in Section 6, and placed on a common calendar from 1 January 2018 to 26 September 2026. The average demands of the areas sum to 1.66 TW—North America 455 GW, Europe 339 GW, Asia 839 GW, Oceania 21 GW and

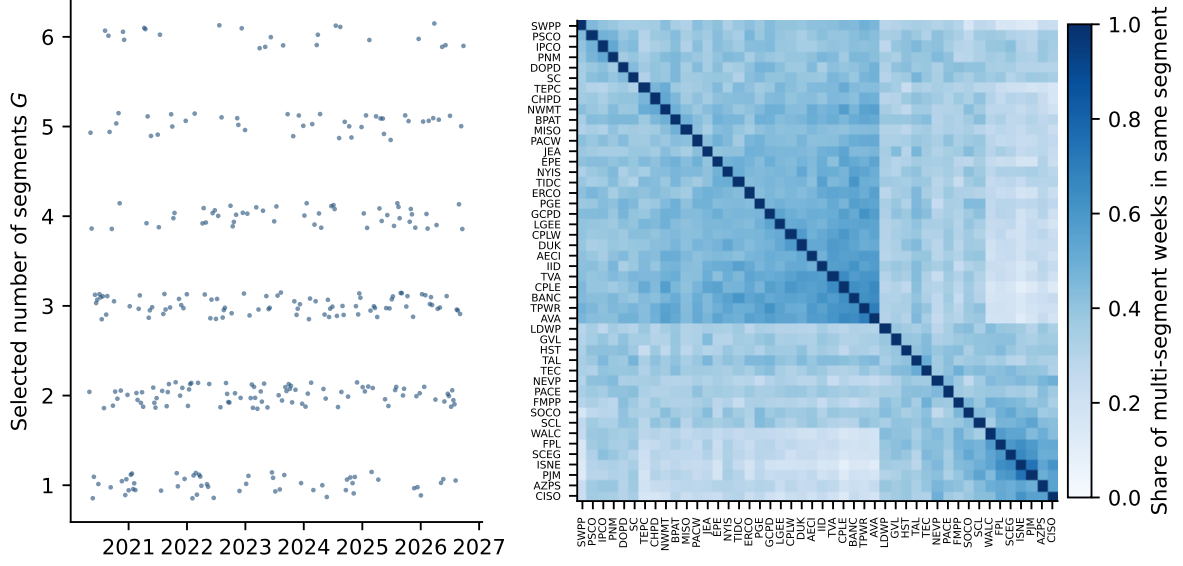


Figure 2: Left: number of segments selected by GDMA in each weekly re-estimation ( $W = 28$ ). Right: among the weeks with  $G \geq 2$ , share of weeks in which two BAs belong to the same segment, ordered by average-linkage clustering.

Africa 7 GW—but the areas are observed in different periods (the Chinese provinces only in 2018), so this is not the demand served at any one time.

The panel is deliberately unbalanced. Systems in Asia and Africa enter and leave, and several have only one or two years of data, which is typical of the data situation in emerging economies; an area contributes to estimation and evaluation only on days with data. Public hourly data for other large African and Asian systems (South Africa, India, Japan, South Korea) exist but could not be retrieved without registration or were not reachable from our computing environment; we discuss this limitation in Section 9.

## 7.2. Candidates, costs and competitors

Every area receives eleven candidate forecasts: the seven of Section 6; a zero-shot forecast from the pretrained time-series foundation model Chronos-Bolt (the small model; Ansari et al., 2024), which receives the 28 days of hourly demand known at the decision time, forecasts the next 48 hours and contributes the last 24 of them, without any fine-tuning; and three weather-aware global LightGBM models. Chronos was pretrained on public time series that include electricity-load data, so some of our series or periods may have been part of its training corpus; its forecasts should be read with this in mind. Outside the United States candidate 0, the *reference forecaster*, is the global LightGBM forecast; it is a strong model-based rule, not the best single candidate, which is reported separately as the per-area best single candidate chosen in each window. The temperature model adds to the LightGBM features the hourly 2-m temperature of the operating day, its daily mean, minimum and maximum, heating and cooling degrees (base 18°C) and the previous day’s mean temperature. Temperatures come from NASA POWER (NASA Langley Research Center, 2026), which provides MERRA-2-based hourly data for any location without registration; each area is represented by the average of one to five load centres (169 locations in total, for example Dallas, Houston and San Antonio for ERCOT, or Paris, Lyon, Marseille and Lille for France). Realised temperature would be a perfect weather forecast, an *ex post* forecast in the terminology of Hong and Fan (2016). The

Table 5: System cost during extreme events relative to the official-forecast policy ( $W = 28$ ): Winter Storm Uri (10–20 February 2021), Winter Storm Elliott (21–26 December 2022, the window of the FERC–NERC inquiry) and the summer heat of 15 July–15 August 2023.

| Method                        | Uri          | Elliott      | Heat wave 2023 |
|-------------------------------|--------------|--------------|----------------|
| Official forecast             | 1.000        | <b>1.000</b> | 1.000          |
| Equal weights                 | 1.292        | 2.984        | 1.501          |
| Best single (per BA)          | 0.982        | 1.204        | 0.828          |
| Pooled DF weights             | 0.955        | 1.303        | 0.816          |
| Per-BA DF weights             | 0.941        | 1.189        | 0.836          |
| Shrinkage to pooled           | 0.932        | 1.189        | 0.834          |
| Two-step $k$ -means           | 0.931        | 1.180        | 0.811          |
| Forecast-then-commit, per BA  | 1.006        | 1.512        | 0.829          |
| Forecast-then-commit, grouped | 0.974        | 1.369        | <b>0.807</b>   |
| GDMA                          | 0.951        | 1.275        | 0.818          |
| Quantile-regression comb.     | 1.011        | 1.488        | 0.821          |
| GDMA, minimum hold-out rule   | 0.935        | 1.169        | 0.825          |
| <b>Soft GDMA</b>              | <b>0.931</b> | 1.205        | 0.827          |

weather models therefore use *archived day-ahead forecasts* of NOAA’s Global Forecast System ([NOAA National Centers for Environmental Prediction, 2026b](#)): for day  $d$  we take the run initialised at 00 UTC on day  $d - 1$  and its 2-m temperature forecasts for lead times of 24 to 48 hours, interpolated to the load centres and to hourly resolution. This run becomes available at about 04–05 UTC, before a 10:00-local gate closure in the Americas, Europe and Algeria but after it in East Asia and Australia, so the weather models use numerical weather forecasts only in the first group of areas. Only the required fields are downloaded from each 1-degree forecast file, by byte range (18,846 temperature fields, 99.8% retrieved). The archive we use, NOAA’s open-data archive of 1-degree GFS files, starts in January 2021 (longer archives exist, for example at NCEI and in the GEFSv12 reforecast, but were not used), so the weather models have forecasts from 2021 onwards and fall back on their other features before. The previous day’s temperature is not yet observed when the commitment is made; it is taken from the GFS forecast for day  $d - 1$  issued on day  $d - 2$ . Against realised temperatures the GFS forecasts have a root-mean-squared error of 2.3°C (1.9°C in Europe, 2.2°C in North America, 2.7°C in Asia). A version of the temperature model with realised temperatures is reported as an upper bound on the value of temperature information. The *multi-variable model* adds, from the same GFS runs and lead times, the forecast 2-m relative humidity, 10-m wind speed and downward short-wave radiation at the surface (99.8% of the files from 2021 retrieved), and from the GEFSv12 ensemble ([Zhou et al., 2022](#); [NOAA National Centers for Environmental Prediction, 2026a](#)) the ensemble standard deviation of 2-m temperature at 0.5 degrees (available for every day from October 2020), each hourly and as daily means. The spread measures how uncertain tomorrow’s weather is, which a point forecast cannot convey, so the third weather candidate uses it for the commitment margin rather than for the point forecast: it takes the multi-variable forecast and multiplies its quantile margin by the ratio of the operating day’s spread to its mean over the previous 28 days, clipped to  $[0.5, 3]$  (*ensemble margin*). The rule was motivated by the gap to the operators’ forecasts in Winter Storm Elliott; its constants were chosen once and not tuned, but since it was designed with that event in view, and uses a raw, uncalibrated ensemble spread from a

Table 6: Sensitivity to the cost asymmetry ( $W = 28$ ): system cost relative to the official-forecast policy when every BA has the same critical ratio  $\tau$ , and under the calibrated heterogeneous ratios.

| Method                        | $\tau = 0.5$  | $\tau = 0.8$  | $\tau = 0.9$  | $\tau = 0.95$ | Calibrated    |
|-------------------------------|---------------|---------------|---------------|---------------|---------------|
| Official forecast             | 1.0000        | 1.0000        | 1.0000        | 1.0000        | 1.0000        |
| Equal weights                 | 1.5702        | 1.5975        | 1.6418        | 1.6623        | 1.6902        |
| Best single (per BA)          | 0.8459        | 0.8240        | 0.8346        | 0.8562        | 0.8585        |
| Pooled DF weights             | 0.8351        | 0.8145        | 0.8246        | 0.8424        | 0.8446        |
| Per-BA DF weights             | 0.8349        | 0.8134        | 0.8226        | 0.8389        | 0.8422        |
| Shrinkage to pooled           | 0.8307        | 0.8100        | 0.8191        | 0.8345        | 0.8385        |
| Two-step $k$ -means           | <b>0.8268</b> | <b>0.8055</b> | <b>0.8142</b> | <b>0.8294</b> | <b>0.8348</b> |
| Forecast-then-commit, per BA  | 0.8470        | 0.8266        | 0.8420        | 0.8687        | 0.8673        |
| Forecast-then-commit, grouped | 0.8423        | 0.8213        | 0.8335        | 0.8561        | 0.8548        |
| GDMA                          | 0.8297        | 0.8092        | 0.8171        | 0.8337        | 0.8356        |
| Quantile-regression comb.     | 0.8457        | 0.8268        | 0.8495        | 0.8904        | 0.8834        |
| GDMA, minimum hold-out rule   | 0.8317        | 0.8096        | 0.8187        | 0.8362        | 0.8383        |
| <b>Soft GDMA</b>              | 0.8303        | 0.8088        | 0.8171        | 0.8326        | 0.8355        |

Table 7: The global panel.

| Continent     | Systems                              | Areas | Period    | Mean demand |
|---------------|--------------------------------------|-------|-----------|-------------|
| North America | U.S. balancing authorities (EIA-930) | 46    | 2019–2026 | 455 GW      |
| Europe        | National systems (Energy-Charts)     | 33    | 2018–2026 | 339 GW      |
| Oceania       | NEM regions, Australia (AEMO)        | 5     | 2018–2026 | 21 GW       |
| Asia          | China (31 provinces)                 | 31    | 2018      |             |
|               | Taiwan (4 grid areas)                | 4     | 2018–2022 | 839 GW      |
|               | Thailand (5 regions)                 | 5     | 2023–2024 |             |
| Africa        | Algeria (Sonelgaz)                   | 1     | 2018–2020 | 7 GW        |
| Total         |                                      | 125   | 2018–2026 | 1.66 TW     |

single archive, it should be read as exploratory. On an average day the spread says little about the size of the error (rank correlation 0.04 across area-days, median 0.08 within areas), but it is large precisely on days of unusual weather. Over the evaluation period the mean absolute errors, scaled by mean demand, are 3.5% for the temperature and multi-variable models (3.5% with exact temperatures), 3.6% for LightGBM, 3.8% for Chronos and for the hourly regression, 4.5% for the official/LightGBM candidate and 6–8% for the naive rules; the foundation model is thus competitive with a model trained on the data themselves.

Critical ratios follow the flexibility rule of Section 6 wherever interchange is measured (the United States, 31 European countries, and the isolated island grid of Taiwan, whose flexibility is zero); the remaining areas receive the midpoint  $\tau = 0.875$ . Section 6.7 and Table D.22 in the appendix show that a common  $\tau$  gives the same ranking.

Besides the benchmarks of Section 6 we add a deep-learning competitor, a *decision-focused neural gate*. It is a two-hidden-layer perceptron (NGPARAMS parameters with eleven candidates) that maps each area’s features at the weekly origin—each candidate’s recent decision cost and forecast error over the last 7 and 28 known days, the area’s critical ratio, recent mean demand, share of days with data and continent, and the season—into

Table 8: Out-of-sample commitment cost on five continents ( $W = 28$  days, May 2018 to September 2026, eleven candidates including the three weather models fed GFS and GEFS forecasts), relative to the reference forecaster. Bold: lowest in the column. Stars: Diebold–Mariano test of equal daily cost against soft GDMA (Newey–West, 7 lags); \*, \*\*, \*\*\*:  $p < 0.10, 0.05, 0.01$ . †: member of the 90% model confidence set (world daily cost, block bootstrap).

| Method                            | World            | North America  | Europe       | Oceania         | Asia            | Africa        |
|-----------------------------------|------------------|----------------|--------------|-----------------|-----------------|---------------|
| Reference forecaster <sup>a</sup> | 1.000***         | 1.000***       | 1.000***     | 1.000***        | 1.000           | 1.000***      |
| Equal weights                     | 1.217***         | 1.437***       | 1.062***     | 0.993***        | 1.033***        | 0.977***      |
| Best single (per area)            | 0.923***         | 0.872***       | 0.946***     | 1.007***        | 1.018***        | 0.904         |
| Pooled DF weights                 | 0.902***         | 0.874***       | 0.915**      | 0.982**         | <b>0.929***</b> | 0.916         |
| Per-area DF weights               | 0.895***         | 0.848***       | 0.913***     | 0.988***        | 0.985***        | 0.920**       |
| Shrinkage to pooled               | 0.888***         | 0.844***       | 0.906***     | 0.983***        | 0.963           | 0.910         |
| Two-step $k$ -means               | 0.884†           | 0.849***       | 0.897        | 0.975           | 0.941***        | 0.888         |
| Quantile-regression comb.         | 1.016***         | 1.005          | 0.989***     | 0.986**         | 1.222**         | 1.030***      |
| Forecast-then-commit, per area    | 0.988**          | 0.985          | 0.934***     | 0.987***        | 1.286**         | 0.929         |
| Forecast-then-commit, grouped     | 0.966*           | 0.978          | 0.899        | 0.978           | 1.249*          | 0.914         |
| Neural gate (deep MoE)            | 0.956***         | 1.007***       | <b>0.896</b> | <b>0.964***</b> | 1.008           | 0.899         |
| GDMA                              | 0.884†           | 0.843***       | 0.901        | 0.972**         | 0.951***        | <b>0.879*</b> |
| <b>Soft GDMA</b>                  | 0.884†           | 0.839          | 0.902        | 0.975           | 0.965           | 0.895         |
| Fair soft GDMA                    | <b>0.881***†</b> | <b>0.837**</b> | 0.897***     | 0.974           | 0.963           | 0.892         |
| Grid areas                        | 125              | 46             | 33           | 5               | 40              | 1             |

<sup>a</sup> The operator’s official forecast where one is published (United States), otherwise the global LightGBM forecast.

combination weights through a softmax, as in mixture-of-experts model averaging (Gong et al., 2026). It is trained end to end on the same smoothed newsvendor cost, on the up to 52 most recent weeks whose outcomes are fully observed, and re-trained every four weeks; the four most recent of these weeks serve for early stopping, and the weights of two independently initialised networks are averaged. We did not search over architectures, so the gate represents a reasonably trained but untuned neural benchmark. Unlike GDMA, it can let weights vary continuously with observed conditions; like GDMA, it only uses information available at the time of each decision.

### 7.3. Results

Table 8 shows that the main findings of the U.S. analysis carry over to the world panel, with smaller savings. Worldwide, soft GDMA lowers the commitment cost by 11.6% relative to the reference forecaster (the operators’ official forecast in the United States, a global LightGBM forecast elsewhere) and by 4.2 percentage points relative to the best single candidate chosen in each window; its fair variant, discussed below, lowers it by 11.9%. At the 90% level the model confidence set (Hansen et al., 2011) contains four methods—two-step clustering, GDMA, soft GDMA and fair soft GDMA—and excludes shrinkage towards pooled weights only narrowly ( $p = 0.098$ ), per-area and pooled weights equally narrowly, and the neural gate, the quantile-regression combination and the best single candidate more clearly. Within the confidence set the differences are below 0.4% and not statistically distinguishable; fair soft GDMA is the cheapest in total (0.881), significantly so against soft GDMA. By continent the cheapest method is a GDMA variant in North America and Africa; in Europe and Oceania the neural gate is nominally cheapest (0.896 and 0.964, against 0.897 for fair soft GDMA in Europe and 0.972 for GDMA in Oceania), and in Asia pooled weights are cheapest (0.929 against 0.963–0.965), because

Table 9: World commitment cost relative to the reference forecaster for four candidate sets ( $W = 28$ ): eight candidates without weather; nine with the temperature model fed archived GFS day-ahead forecasts; eleven with, in addition, the multi-variable model and its ensemble-margin version (main specification); and nine with the temperature model fed realised temperatures (upper bound on temperature information).

| Method                            | No weather | GFS temperature | GFS/GEFS multi-variable (main) | Realised temperature |
|-----------------------------------|------------|-----------------|--------------------------------|----------------------|
| Reference forecaster <sup>a</sup> | 1.000      | 1.000           | 1.000                          | 1.000                |
| Equal weights                     | 1.380      | 1.312           | 1.217                          | 1.308                |
| Best single (per area)            | 0.942      | 0.929           | 0.923                          | 0.915                |
| Pooled DF weights                 | 0.929      | 0.908           | 0.902                          | 0.894                |
| Per-area DF weights               | 0.918      | 0.903           | 0.895                          | 0.886                |
| Shrinkage to pooled               | 0.911      | 0.896           | 0.888                          | 0.879                |
| Two-step $k$ -means               | 0.904      | 0.889           | 0.884                          | 0.875                |
| Quantile-regression comb.         | 1.038      | 1.022           | 1.016                          | 1.005                |
| Forecast-then-commit, per area    | 1.011      | 0.994           | 0.988                          | 0.977                |
| Forecast-then-commit, grouped     | 0.988      | 0.971           | 0.966                          | 0.957                |
| Neural gate (deep MoE)            | 0.968      | 0.958           | 0.956                          | —                    |
| GDMA                              | 0.905      | 0.889           | 0.884                          | 0.875                |
| <b>Soft GDMA</b>                  | 0.906      | 0.892           | 0.884                          | 0.876                |
| Fair soft GDMA                    | 0.905      | 0.887           | 0.881                          | 0.871                |

most Asian areas have a single year or less of data.

Three comparisons stand out. First, decision-focused combination on the simplex dominates forecast-then-commit and the unconstrained quantile-regression combination: combining the same forecasts for accuracy and adding a quantile margin saves only 1–3% worldwide, and the quantile-regression combination is more expensive than the reference forecaster (1.016) because, with eleven highly collinear candidates and a few weeks of data, freeing the weights from the simplex costs more in variance than it gains in flexibility; both are particularly poor in Asia, where histories are short. Second, a neural gate trained with early stopping and averaged over two initialisations is clearly worse than the simple decision-focused rules (0.956 against 0.884 worldwide), although it is competitive in Europe and Oceania. We did not search over architectures, so this should be read as evidence that flexibility alone does not solve the problem, not as a statement about the best achievable neural design. Third, better candidates and better combination are complements. On their own the temperature model costs 0.965 of the reference forecaster, the multi-variable model 0.960 and its ensemble-margin version 0.981, the wider margins costing money on ordinary days; the global LightGBM model costs 0.988 and the zero-shot foundation model 1.434, which is much worse than with a one-day-ahead information set because it must now forecast two days ahead. Adding the temperature model to the candidate set lowers the cost of soft GDMA from 0.906 to 0.892, and adding the two multi-variable candidates lowers it further to 0.884 (Table 9); with realised temperatures in place of GFS forecasts in the temperature model, soft GDMA costs 0.876. The ranking of the methods is the same for every candidate set, so better weather information helps all methods alike. The ranking is also similar across estimation windows and critical ratios (Tables D.21 and D.22 in the appendix).

*Extreme events.* The U.S. analysis (Table 5) showed that during Winter Storm Elliott, the abrupt cold snap of 21–26 December 2022 (FERC, NERC and Regional Entity Staff,

Table 10: Winter Storm Elliott (21–26 December 2022, 46 U.S. balancing authorities): commitment cost relative to the operators’ official day-ahead forecasts, by candidate set ( $W = 28$ ), and in the main set without the BA-days on which load was shed.

|                                        | No<br>weather | GFS<br>temperature | GFS/GEFS<br>multi-variable | Realised<br>temperature | Main set,<br>no shed days |
|----------------------------------------|---------------|--------------------|----------------------------|-------------------------|---------------------------|
| <i>Combination rules</i>               |               |                    |                            |                         |                           |
| <b>Soft GDMA</b>                       | 1.200         | 1.147              | 1.095                      | 1.111                   | 1.084                     |
| Fair soft GDMA                         | 1.176         | 1.084              | 1.112                      | 1.091                   | 1.103                     |
| GDMA                                   | 1.250         | 1.216              | 1.071                      | 1.121                   | 1.056                     |
| Two-step $k$ -means                    | 1.197         | 1.143              | 1.128                      | 1.105                   | 1.123                     |
| Shrinkage to pooled                    | 1.198         | 1.136              | 1.138                      | 1.121                   | 1.128                     |
| Per-area DF weights                    | 1.169         | 1.127              | 1.138                      | 1.116                   | 1.128                     |
| Pooled DF weights                      | 1.299         | 1.177              | 1.169                      | 1.159                   | 1.143                     |
| Best single (per area)                 | 1.239         | 1.144              | 1.136                      | 1.131                   | 1.100                     |
| Quantile-regression comb.              | 1.361         | 1.354              | 1.367                      | 1.345                   | 1.351                     |
| Forecast-then-commit, grouped          | 1.395         | 1.170              | 1.171                      | 1.198                   | 1.152                     |
| <i>Weather candidates on their own</i> |               |                    |                            |                         |                           |
| Temperature model                      | –             | 1.100              | 1.100                      | 1.112                   | –                         |
| Multi-variable model                   | –             | –                  | 1.074                      | –                       | –                         |
| Multi-variable, ensemble margin        | –             | –                  | 0.929                      | –                       | –                         |

2023), the operators’ own forecasts were better than every combination. Table 10 shows how much of that gap is a matter of weather information. Without weather candidates soft GDMA is 20.0% more expensive than the official forecasts over the six days; temperature forecasts from GFS reduce this to 14.7%, and the multi-variable candidates, which add humidity, wind, radiation and the ensemble spread, to 9.5%. With realised temperatures in the temperature model the figure is 11.1%, so the multi-variable set is not better than a perfect temperature forecast in any structural sense: the six days are too few to rank candidate sets reliably, and rankings among the combination rules also change with the set. Excluding the BA-days on which TVA and Duke shed load (FERC, NERC and Regional Entity Staff, 2023), which understate true demand, soft GDMA is 8.4% more expensive. The only rule that beats the operators during the storm is a single candidate: the multi-variable model with the ensemble margin, which is 7% cheaper (0.929), because the GEFS spread rose on 23 December, the costliest day, and widened the margin that day by a median factor of 1.4. The rule was designed after the gap in Elliott was known, precisely to address it, so this result is exploratory: it is a single candidate, evaluated on the single event that motivated it; over the whole period it is the most expensive of the three weather candidates (0.981), and the combination rules, whose weights are learned from the previous four weeks, do not give it enough weight to exploit it. During Winter Storm Uri, soft GDMA is 11–13% cheaper than the operators in every candidate set, and 11–13% cheaper also when ERCOT’s, SPP’s and MISO’s load-shed days are excluded.

A single storm is a weak basis for conclusions about extremes, so we also examine all extreme-temperature days by a rule fixed beforehand: the area-days on which the daily mean temperature is below its 2.5% or above its 97.5% quantile over the evaluation period (Table 11). On cold days soft GDMA is 12% cheaper than the reference forecaster worldwide (90% interval 0.83–0.93, block bootstrap over days) and on hot days 19% cheaper (0.79–0.84); in the United States 10% and 25% cheaper. Elliott is therefore not typical of extreme weather in general; it is an abrupt cold snap accompanied by large forced outages, and the loss of the combination rules in that event reflects weights that

Table 11: Commitment cost relative to the reference forecaster on extreme-temperature area-days ( $W = 28$ ): days on which an area’s daily mean temperature is below its 2.5% or above its 97.5% quantile. Intervals: moving-block bootstrap over days (7-day blocks, 1,000 replications).

|                                   | World<br>cold | World<br>hot | U.S.<br>cold | U.S.<br>hot  |
|-----------------------------------|---------------|--------------|--------------|--------------|
| Reference forecaster <sup>a</sup> | 1.000         | 1.000        | 1.000        | 1.000        |
| Equal weights                     | 1.239         | 0.975        | 1.620        | 1.066        |
| Best single (per area)            | 0.952         | 0.879        | 0.962        | 0.810        |
| Pooled DF weights                 | 0.903         | 0.827        | 0.975        | 0.776        |
| Per-area DF weights               | 0.901         | 0.822        | 0.923        | 0.754        |
| Shrinkage to pooled               | 0.889         | 0.817        | 0.912        | 0.751        |
| Two-step $k$ -means               | 0.876         | 0.827        | 0.913        | 0.768        |
| Quantile-regression comb.         | 1.138         | 0.971        | 1.144        | 0.864        |
| Forecast-then-commit, per area    | 1.048         | 0.910        | 1.106        | 0.820        |
| Forecast-then-commit, grouped     | 0.954         | 0.885        | 1.012        | 0.808        |
| GDMA                              | 0.877         | 0.818        | 0.905        | 0.748        |
| <b>Soft GDMA</b>                  | 0.879         | 0.814        | 0.898        | 0.747        |
| Fair soft GDMA                    | 0.884         | 0.815        | 0.904        | 0.752        |
| Soft GDMA / reference, 90% CI     | [0.83, 0.93]  | [0.79, 0.84] | [0.86, 0.94] | [0.71, 0.79] |
| Soft GDMA / best single, 90% CI   | [0.90, 0.94]  | [0.91, 0.94] | [0.91, 0.96] | [0.89, 0.95] |
| Area-days                         | 9,624         | 9,623        | 3,542        | 3,540        |

<sup>a</sup> As in Table 8.

cannot anticipate a regime change.

*Tail risk.* The linear cost model hides the tail, so Table 12 reports the average of the costliest 5% of days ( $\text{CVaR}_{95}$ ) and the costliest week of the world’s daily commitment cost. Soft GDMA’s costliest week is 16% cheaper than the reference forecaster’s (0.840) and its  $\text{CVaR}_{95}$  is 9.5% lower (0.905), so its saving is not bought with a heavier tail. The forecast-then-commit and quantile-regression rules fail in the tail: their  $\text{CVaR}_{95}$  is 55–63% higher than the reference forecaster’s and their costliest week is nine times as costly, mostly in the first week of January 2019, when the 46 U.S. balancing authorities enter the panel with a few days of data; even without each area’s first eight weeks their costliest week is more than twice the reference’s. Rules that estimate weights on the simplex, with a shared structure across areas, do not fail in this way.

*New and data-poor systems.* Fifty-two areas enter the panel after the start of the evaluation: the 46 U.S. balancing authorities in January 2019, the five Thai regions in January 2023 and one European system. During their first eight weeks each of them has only a few weeks of history (Table 13). No combination does well in this phase: all of them except the quantile-regression combination (0.976) are more expensive than the reference forecaster during the first eight weeks. Pooled weights, which impose the rule learned from other systems, are 61% more expensive, forecast-then-commit 39–40% more expensive, and the neural gate, whose features are not yet informative, 2.6 times as expensive. Plain GDMA (1.053) and fair soft GDMA (1.058) are the closest to the reference among the rules that share structure across areas, followed by soft GDMA and the best single candidate (1.069, each); per-area weights, which rely on the area’s own few days, and shrinkage are 11% more expensive. Two-step clustering (1.170) does less well for entering areas than for the established ones, because it clusters weights that are themselves estimated from a handful of days. The quantile-regression combination is better than the reference for entering

Table 12: Tail risk of the world’s daily commitment cost relative to the reference forecaster ( $W = 28$ ): total, average of the costliest 5% of days (CVaR<sub>95</sub>) and costliest week, for the calibrated critical ratios, without each area’s first eight weeks, and with a scarcity-level critical ratio of 0.99 for every area.

| Method                            | Calibrated $\tau_i$ (main) |                          |            | Without first 8 weeks of each area |            | $\tau_i = 0.99$ for every area |                          |            |
|-----------------------------------|----------------------------|--------------------------|------------|------------------------------------|------------|--------------------------------|--------------------------|------------|
|                                   | Total                      | CVaR <sub>95</sub> daily | Worst week | CVaR <sub>95</sub> daily           | Worst week | Total                          | CVaR <sub>95</sub> daily | Worst week |
| Reference forecaster <sup>a</sup> | 1.000                      | 1.000                    | 1.000      | 1.000                              | 1.000      | –                              | –                        | –          |
| Equal weights                     | 1.217                      | 1.136                    | 1.068      | 1.114                              | 1.019      | –                              | –                        | –          |
| Best single (per area)            | 0.923                      | 0.981                    | 0.946      | 0.971                              | 0.905      | –                              | –                        | –          |
| Pooled DF weights                 | 0.902                      | 0.914                    | 0.919      | 0.901                              | 0.830      | –                              | –                        | –          |
| Per-area DF weights               | 0.895                      | 0.920                    | 0.829      | 0.908                              | 0.845      | –                              | –                        | –          |
| Shrinkage to pooled               | 0.888                      | 0.902                    | 0.845      | 0.893                              | 0.805      | –                              | –                        | –          |
| Two-step $k$ -means               | 0.884                      | 0.892                    | 0.853      | 0.884                              | 0.785      | –                              | –                        | –          |
| Quantile-regression comb.         | 1.016                      | 1.634                    | 8.955      | 1.152                              | 2.316      | –                              | –                        | –          |
| Forecast-then-commit, per area    | 0.988                      | 1.603                    | 8.968      | 1.117                              | 2.313      | –                              | –                        | –          |
| Forecast-then-commit, grouped     | 0.966                      | 1.554                    | 8.995      | 1.067                              | 2.274      | –                              | –                        | –          |
| Neural gate (deep MoE)            | 0.956                      | 0.943                    | 0.996      | 0.859                              | 0.794      | –                              | –                        | –          |
| GDMA                              | 0.884                      | 0.902                    | 0.866      | 0.893                              | 0.804      | –                              | –                        | –          |
| <b>Soft GDMA</b>                  | 0.884                      | 0.905                    | 0.840      | 0.895                              | 0.820      | –                              | –                        | –          |
| Fair soft GDMA                    | 0.881                      | 0.898                    | 0.848      | 0.888                              | 0.826      | –                              | –                        | –          |

areas but worse for established ones (1.016); we have no explanation for this reversal and the entry sample is small. In the U.S. part of the sample, where an official forecast exists, an operator adopting any of these rules would do well to keep committing on it for the first weeks of a new unit.

*Segments across continents.* GDMA uses more than one segment in 89% of the weekly re-estimations (two segments in 20%, three in 22%, four in 21%), much more often than in the U.S. panel, and soft GDMA shrinks towards the segment rule with an average intensity of  $\kappa = 0.39$ . Segments are not continents. Among weeks with several segments, two systems on the same continent share a segment only 37–49% of the time (North America 46%, Europe 49%, Oceania 42%, Asia 37%), European and African systems are grouped together 52% of the time, and North American areas are grouped with European or Asian ones only 19–21% of the time. The segmentation therefore does not follow geography; it is consistent with differences in which forecasting systems work—notably whether an operator’s own forecast is available and informative—which a planner could not have specified in advance.

#### 7.4. Who benefits? A fairness audit

Table 14 asks how the gains are distributed, relative to the reference forecaster and relative to each area’s own best single candidate. Against the reference forecaster the picture is mixed: even the best rules make 17–21% of the areas worse off, because the reference is often a strong forecast (the operator’s own, or a model forecast that is also one of the candidates), and rules that impose a common rule or learn a flexible global map do worse: the neural gate makes 34% of the areas worse off, forecast-then-commit 41–45% and the quantile-regression combination 58%. Against each area’s own best single candidate the groups separate more clearly: soft GDMA makes only 6% of the areas worse off and its fair variant 3%, whereas pooled weights make 24% worse off, per-area weights and two-step clustering 14%, the neural gate 42%, forecast-then-commit 52–60% and the quantile-regression combination 86%. The median area in the worst-served continent is 2.2–2.8% better off than under the reference under the GDMA variants and the clustering

Table 13: Commitment cost of areas entering the panel, relative to the reference forecaster ( $W = 28$ , eleven candidates).

| Method                            | First 8 weeks after entry <sup>b</sup> | All other weeks |
|-----------------------------------|----------------------------------------|-----------------|
| Reference forecaster <sup>a</sup> | 1.000                                  | 1.000           |
| Equal weights                     | 2.477                                  | 1.206           |
| Best single (per area)            | 1.069                                  | 0.922           |
| Pooled DF weights                 | 1.614                                  | 0.896           |
| Per-area DF weights               | 1.111                                  | 0.893           |
| Shrinkage to pooled               | 1.115                                  | 0.886           |
| Two-step $k$ -means               | 1.170                                  | 0.881           |
| Quantile-regression comb.         | 0.976                                  | 1.016           |
| Forecast-then-commit, per area    | 1.394                                  | 0.985           |
| Forecast-then-commit, grouped     | 1.400                                  | 0.962           |
| Neural gate (deep MoE)            | 2.599                                  | 0.942           |
| GDMA                              | 1.053                                  | 0.882           |
| <b>Soft GDMA</b>                  | 1.069                                  | 0.882           |
| Fair soft GDMA                    | 1.058                                  | 0.879           |
| Grid areas entering               | 52                                     |                 |

<sup>b</sup> The first eight weeks after an area has at least one day of data in the estimation window.

rules, and worse off under the neural gate (1.010), forecast-then-commit (1.047–1.053) and the quantile-regression combination (1.127). A large gap remains between data-rich and data-poor systems: areas with at least two years of data save 12% under soft GDMA, areas with less than two years (the Chinese provinces and Algeria) only 4%, and per-area weights save almost nothing for the latter (0.983); only pooled weights, which impose the rule of other systems, do nearly as well for data-poor as for data-rich areas (0.922 against 0.901), at a higher cost for the rich ones.

Fair soft GDMA, which counts every area’s relative improvement equally, is also the cheapest method in total (0.881 against 0.884, significant at the 1% level), although within the model confidence set and economically negligible. So in this panel fairness in the sense of the criterion costs essentially nothing in total cost, and shifts the distribution: the share of areas worse off than their own best candidate halves (6% to 3%), the 10th percentile of the ratio falls to 0.840 and its 90th to 1.022. Weighting by relative cost prevents the few largest systems from dictating the segment rules, which is consistent with the segment-structure argument of Section 4, under which fair and efficient criteria share their minimiser; we cannot tell from these data whether segments are exact.

*Income..* Fairness at this scale is also a question of income. We attach to every grid area the GDP per capita at purchasing-power parity of its country and its World Bank income group (World Bank, 2026), with Taiwan from the IMF World Economic Outlook (International Monetary Fund, 2026); 83 areas are in high-income and 42 in upper-middle-income countries. For the U.S. balancing authorities we also compute the per-capita personal income of the states they serve (U.S. Bureau of Economic Analysis, 2026), weighted by approximate service-territory shares. Table 15 reports the results.

Across the world, areas in upper-middle-income countries save less than areas in high-income countries under every decision-focused simplex method (soft GDMA 4.9% against 12.2%), but this gap largely coincides with the data gap above, because the upper-middle-income group is dominated by the Chinese provinces with a single year of data. Across individual areas, the rank correlation between an area’s cost ratio and its GDP per capita

Table 14: Distribution of the benefits across the 125 grid areas ( $W = 28$ , eleven candidates): percentiles of each area’s cost relative to the reference forecaster, share of areas made worse off than under the reference and than under their own best single candidate, median ratio in the worst-served continent, and cost ratios for data-rich and data-poor areas.

| Method                            | Per-area cost ratio |        |       | Areas harmed vs |             | Worst                  | Data-rich | Data-poor          |
|-----------------------------------|---------------------|--------|-------|-----------------|-------------|------------------------|-----------|--------------------|
|                                   | P10                 | Median | P90   | ref.            | best single | continent <sup>d</sup> | areas     | areas <sup>e</sup> |
| Reference forecaster <sup>a</sup> | 1.000               | 1.000  | 1.000 | 0.00            | 0.62        | 1.000                  | 1.000     | 1.000              |
| Equal weights                     | 0.963               | 1.050  | 1.652 | 0.71            | 0.79        | 1.337                  | 1.231     | 1.024              |
| Best single (per area)            | 0.876               | 0.976  | 1.047 | 0.38            | –           | 1.010                  | 0.917     | 1.006              |
| Pooled DF weights                 | 0.856               | 0.935  | 1.037 | 0.19            | 0.24        | 0.984                  | 0.901     | 0.922              |
| Per-area DF weights               | 0.852               | 0.952  | 1.034 | 0.27            | 0.14        | 0.991                  | 0.888     | 0.983              |
| Shrinkage to pooled               | 0.848               | 0.941  | 1.027 | 0.21            | 0.09        | 0.987                  | 0.883     | 0.958              |
| Two-step $k$ -means               | 0.846               | 0.927  | 1.020 | 0.18            | 0.14        | 0.974                  | 0.880     | 0.936              |
| Quantile-regression comb.         | 0.917               | 1.032  | 1.221 | 0.58            | 0.86        | 1.127                  | 1.010     | 1.097              |
| Forecast-then-commit, per area    | 0.887               | 0.985  | 1.232 | 0.45            | 0.60        | 1.047                  | 0.982     | 1.064              |
| Forecast-then-commit, grouped     | 0.868               | 0.972  | 1.214 | 0.41            | 0.52        | 1.053                  | 0.962     | 1.028              |
| Neural gate (deep MoE)            | 0.872               | 0.956  | 1.109 | 0.34            | 0.42        | 1.010                  | 0.952     | 1.008              |
| GDMA                              | 0.854               | 0.934  | 1.015 | 0.17            | 0.11        | 0.972                  | 0.879     | 0.947              |
| <b>Soft GDMA</b>                  | 0.843               | 0.943  | 1.028 | 0.20            | 0.06        | 0.978                  | 0.878     | 0.964              |
| Fair soft GDMA                    | 0.840               | 0.937  | 1.022 | 0.19            | 0.03        | 0.977                  | 0.875     | 0.960              |
| Areas                             | 125                 |        |       |                 |             |                        | 88        | 37                 |

<sup>d</sup> Highest continent-level median of the per-area ratios. <sup>e</sup> Areas with less than two years of evaluation data (the 31 Chinese provinces and Algeria).

is  $-0.12$  for soft GDMA (90% bootstrap interval  $-0.26$  to  $0.03$ ) and  $-0.11$  for its fair variant, so richer areas gain slightly more but the interval includes zero; it is positive and significant for pooled weights (0.19, interval 0.04 to 0.34), which favour poorer areas, and not different from zero for the neural gate. Among the 88 areas with at least two years of data it is positive for each of the methods we checked (0.17 for soft GDMA, 0.18 for its fair variant, 0.23 for pooled weights and 0.39 for the neural gate), that is, richer areas gain *less*. The sign is therefore not stable across subsamples, as one would expect if income itself plays no role and the correlation reflects data length and continent, with which income is confounded. Within the United States, in cost-weighted terms the balancing authorities in the highest third of state income gain more (cost ratios of 0.68 under soft GDMA against 0.93 in the lowest third), but this reflects a few large, high-income systems whose own official forecasts happen to be weak; across balancing authorities the rank correlation with state income is 0.03 (90% interval  $-0.24$  to  $0.31$ ). We therefore find no evidence that GDMA systematically favours richer or poorer systems, but with 125 areas, income measured at the level of countries and states, and strong confounding with data length, the data cannot rule out moderate gradients either.

## 8. Where to install batteries

We now solve the planning problem (4) on the global panel. For each commitment rule we take its out-of-sample hourly shortfalls and surpluses ( $W = 28$ , eleven candidates with GFS and GEFS weather forecasts, decisions taken with the information available at the day-ahead gate closure), place a four-hour battery with 85% round-trip efficiency in every grid area, and compute the annual value of battery power on a grid from 0 to 15% of the area’s mean demand. Costs are converted to money by valuing one MWh of idle committed capacity at an illustrative \$10, so that a shortfall costs between \$40 and \$190 per MWh depending on the area’s critical ratio; all dollar figures scale proportionally with this assumption. The battery is dispatched greedily against the given

Table 15: Benefits by income ( $W = 28$ ): commitment cost relative to the reference forecaster by World Bank income group and, within the United States, for the balancing authorities in the lowest and highest third of state per-capita income; Spearman rank correlations between an area’s cost ratio and income.

| Method                            | All 125 areas |                     |                                       | U.S. balancing authorities |                      |                                           |
|-----------------------------------|---------------|---------------------|---------------------------------------|----------------------------|----------------------|-------------------------------------------|
|                                   | High income   | Upper-middle income | Rank corr. with GDP p.c. <sup>g</sup> | Lowest income third        | Highest income third | Rank corr. with state income <sup>g</sup> |
| Reference forecaster <sup>a</sup> | 1.000         | 1.000               | –                                     | 1.000                      | 1.000                | –                                         |
| Equal weights                     | 1.235         | 1.014               | 0.530                                 | 1.636                      | 1.141                | -0.106                                    |
| Best single (per area)            | 0.917         | 0.992               | -0.126                                | 0.964                      | 0.692                | 0.013                                     |
| Pooled DF weights                 | 0.901         | 0.918               | 0.185                                 | 0.959                      | 0.729                | -0.033                                    |
| Per-area DF weights               | 0.888         | 0.969               | -0.164                                | 0.941                      | 0.680                | 0.015                                     |
| Shrinkage to pooled               | 0.883         | 0.947               | -0.093                                | 0.936                      | 0.680                | 0.009                                     |
| Two-step $k$ -means               | 0.880         | 0.928               | 0.108                                 | 0.938                      | 0.696                | 0.063                                     |
| Quantile-regression comb.         | 1.011         | 1.074               | -0.204                                | 1.128                      | 0.771                | -0.107                                    |
| Forecast-then-commit, per area    | 0.983         | 1.043               | -0.010                                | 1.101                      | 0.768                | -0.122                                    |
| Forecast-then-commit, grouped     | 0.962         | 1.011               | 0.046                                 | 1.089                      | 0.781                | -0.122                                    |
| Neural gate (deep MoE)            | 0.953         | 0.989               | 0.086                                 | 1.128                      | 0.793                | -0.127                                    |
| GDMA                              | 0.879         | 0.938               | 0.025                                 | 0.931                      | 0.689                | 0.045                                     |
| <b>Soft GDMA</b>                  | 0.878         | 0.951               | -0.120                                | 0.927                      | 0.678                | 0.028                                     |
| Fair soft GDMA                    | 0.875         | 0.947               | -0.114                                | 0.927                      | 0.678                | 0.039                                     |
| Areas                             | 83            | 42                  |                                       | 15                         | 16                   |                                           |

<sup>g</sup> A negative correlation means that richer areas have lower cost ratios, i.e. gain more.

commitment path, charging from committed-but-idle capacity and discharging into shortfalls; the commitment itself is not re-optimised with the battery in place, which would lower the commitment and the battery’s value, especially for rules with large shortfalls. The exercise values a single stream—cover for day-ahead commitment errors—and ignores arbitrage, adequacy, renewable firming and network deferral, which dominate real storage decisions. It should be read as an illustration of how the operational rule changes the value and location of storage, conditional on this stream, not as a siting recommendation; and because the areas are observed over different periods, annual values are averages over each area’s own evaluation period.

### 8.1. Batteries do not pay for themselves on forecast errors alone

The first increment of battery power is worth \$31,000–\$45,000 per MW-year in the median area of each continent (depending on the commitment rule), and at most \$73,000 (in the PJM interconnection). Annualised costs of four-hour batteries are well above this range (we use \$100,000–\$200,000 per MW-year), so the cost-optimal battery size for covering day-ahead commitment errors alone is zero in every area and for every commitment rule. Storage is deployed for arbitrage, capacity adequacy and the integration of renewables, and covering commitment errors is a by-product. The relevant planning question is therefore not whether, but *where*, to place the storage that energy-transition programmes already mandate, and how that choice depends on the way operators commit.

### 8.2. Better commitment rules reduce the value of storage

Figure 3 and Table 16 show how the operational rule and storage interact. Without storage, soft GDMA commits the five continents’ capacity for \$11.3 billion a year under the stylised cost model, against \$12.3 billion for the reference forecaster, a difference of \$1.0 billion or 8%. Adding batteries lowers every curve: the reference rule needs 34 GW of optimally sited batteries to reach the cost that soft GDMA achieves without any, forecast-then-commit needs 17 GW and per-area decision-focused weights 5 GW, while pooled

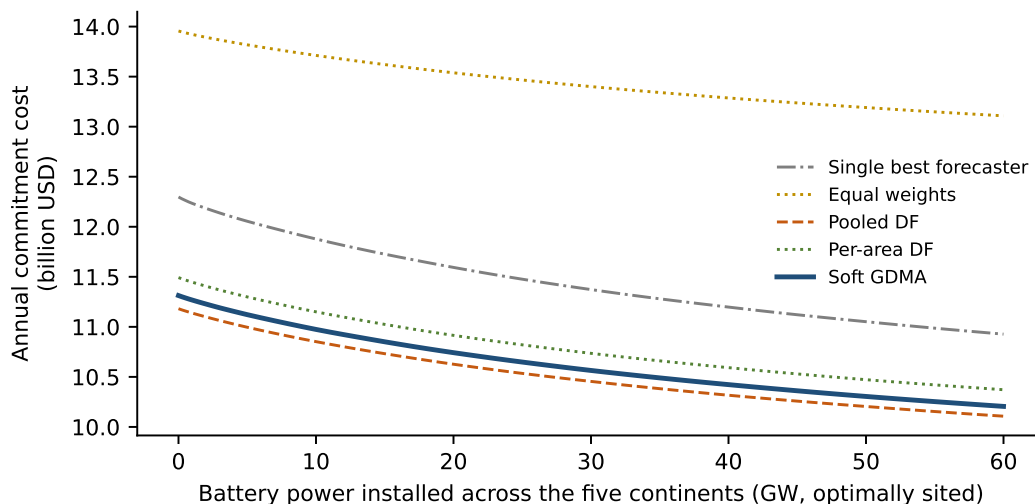


Figure 3: Annual system-wide commitment cost as a function of battery power installed and optimally sited across the 125 grid areas, for five commitment rules. Costs valued at \$10 per MWh of idle committed capacity; annual values average over each area’s own evaluation period.

Table 16: Storage siting on five continents. Cost of each commitment rule without storage, battery power it needs (optimally sited) to reach the cost of soft GDMA without storage, and the optimal siting of a 10 GW storage programme.

| Commitment rule        | Cost without storage (\$bn/yr) | GW to match soft GDMA <sup>c</sup> | Siting of a 10 GW budget (GW) |        |         |      |        |
|------------------------|--------------------------------|------------------------------------|-------------------------------|--------|---------|------|--------|
|                        |                                |                                    | N. Am.                        | Europe | Oceania | Asia | Africa |
| Single best forecaster | 12.30                          | 34                                 | 5.1                           | 1.6    | 0.4     | 2.9  | 0.0    |
| Equal weights          | 13.96                          | > 60                               | 1.5                           | 1.8    | 1.2     | 5.5  | 0.1    |
| Pooled DF              | 11.18                          | 0                                  | 4.1                           | 2.6    | 0.7     | 2.5  | 0.0    |
| Per-area DF            | 11.49                          | 5                                  | 4.3                           | 2.6    | 0.7     | 2.5  | 0.0    |
| Forecast-then-commit   | 11.92                          | 17                                 | 5.0                           | 2.4    | 0.4     | 2.1  | 0.0    |
| GDMA                   | 11.21                          | 0                                  | 4.3                           | 2.6    | 0.7     | 2.4  | 0.0    |
| <b>Soft GDMA</b>       | 11.31                          | –                                  | 4.3                           | 2.6    | 0.7     | 2.5  | 0.0    |

<sup>c</sup> Smallest optimally sited budget, on a grid of 0.25 GW steps up to 5 GW and 1 GW steps up to 60 GW, at which the rule’s cost falls to or below that of soft GDMA without storage.

weights and plain GDMA are as cheap as soft GDMA without storage in this metric. The battery-equivalent should not be read as the value of the better rule. Batteries are credited here only with the commitment errors they absorb, so 34 GW would cost far more than the \$1.0 billion difference in commitment cost; the relevant comparison for an operator is the direct saving. The point is the interaction: the better the commitment rule, the less is left for storage to absorb, and a planner who values storage by today’s errors overstates it.

### 8.3. The siting decision depends on the commitment rule

Table 16 also shows how a 10 GW programme should be distributed. Under soft GDMA, 4.3 GW go to North America, 2.6 GW to Europe, 0.7 GW to Oceania and 2.5 GW to Asia, and Africa receives almost nothing. If the planner instead uses the error profile of the reference forecaster, North America receives 5.1 GW, about 19% more, because the U.S. balancing authorities’ own forecasts leave the largest shortfalls; forecast-then-commit would send 5.0 GW there.

Table 17: Annual value (million USD) of storage programmes of different sizes when operations follow soft GDMA, by the commitment rule whose error profile was used to site the batteries.

| Siting based on        | 1 GW | 5 GW | 10 GW | 20 GW |
|------------------------|------|------|-------|-------|
| Single best forecaster | 43   | 184  | 327   | 560   |
| Per-area DF            | 45   | 191  | 338   | 570   |
| Forecast-then-commit   | 44   | 187  | 333   | 566   |
| <b>Soft GDMA</b>       | 45   | 191  | 338   | 571   |

Table 18: Out-of-sample siting under soft GDMA operations (million USD per year): batteries are sited on the value curves of the first half of the evaluation period and valued on the second half, and compared with siting on the second half itself (infeasible oracle).

| Budget (GW) | Sited on soft GDMA (first half) | Sited on the reference (first half) | Sited on the second half itself (oracle) |
|-------------|---------------------------------|-------------------------------------|------------------------------------------|
| 1           | 44                              | 42                                  | 47                                       |
| 5           | 148                             | 145                                 | 194                                      |
| 10          | 247                             | 232                                 | 336                                      |
| 20          | 391                             | 360                                 | 539                                      |

Table 17 quantifies the cost of siting on the wrong profile, and it is modest: when operations follow soft GDMA, a 1 GW programme sited with the reference rule’s error profile delivers \$43 million a year instead of \$45 million, a 4% loss; the loss is 4% at 5 GW, 3% at 10 GW and 2% at 20 GW. The location of storage therefore matters less than the existence of a good commitment rule, but it is not irrelevant: the difference is a few per cent of the value of the programme, concentrated in North America.

Siting on estimated value curves also carries estimation risk. Table 18 sites batteries using the first half of the evaluation period and values them on the second half: siting on soft GDMA’s own error profile captures about three quarters of the value of the infeasible oracle at 10 GW (\$247 million of \$336 million), and siting on the reference rule’s profile captures 6% less than that at 10 GW and 8% less at 20 GW. Both sources of loss, the choice of rule and the estimation of the profile, are of similar size for programmes of 5–10 GW, and the former does not dominate.

#### 8.4. Fair siting

Efficient siting concentrates batteries where the value in money is highest, and in small programmes some continents receive nothing: with 1 GW, the worst-served continent’s relative benefit is zero (Table 19). Two fair rules respond differently. Floors proportional to each continent’s share of demand, the “fair share” rule often used in practice, raise the worst-served continent’s relative benefit only slightly (from 1.4% to 1.7% of its commitment cost with 10 GW) and give up 3–7% of the programme’s value. Maximin siting, which directs every increment to the continent that has gained least so far, raises it much more—to 2.8% with 10 GW and from zero to 0.3% with 1 GW—at a price of fairness of 7–9%. Fairness in storage siting is therefore not free in this setting: protecting the worst-served continent costs between a thirtieth and a tenth of the programme’s value, and a planner must decide whether that trade is worth making. These figures condition on one value stream and on the continent as the unit of fairness; the within-continent distribution is not examined.

Table 19: Efficient and fair siting of storage programmes under soft GDMA operations: annual value, the relative benefit of the worst-served continent, and the price of fairness (PoF, the share of the efficient programme’s value given up).

| Budget<br>(GW) | Efficient         |                                   | Proportional floors ( $\lambda = 1$ ) |                      |      | Maximin           |                      |      |
|----------------|-------------------|-----------------------------------|---------------------------------------|----------------------|------|-------------------|----------------------|------|
|                | Value<br>(\$m/yr) | Min. rel.<br>benefit <sup>f</sup> | Value<br>(\$m/yr)                     | Min. rel.<br>benefit | PoF  | Value<br>(\$m/yr) | Min. rel.<br>benefit | PoF  |
| 1              | 45                | 0.0%                              | 42                                    | 0.2%                 | 7.3% | 41                | 0.3%                 | 9.0% |
| 5              | 191               | 0.6%                              | 181                                   | 0.9%                 | 5.0% | 178               | 1.5%                 | 6.9% |
| 10             | 338               | 1.4%                              | 325                                   | 1.7%                 | 3.9% | 315               | 2.8%                 | 7.0% |
| 20             | 571               | 3.1%                              | 554                                   | 3.3%                 | 3.0% | 532               | 4.7%                 | 6.7% |

<sup>f</sup> Lowest, across continents, of the value captured divided by the continent’s annual commitment cost without storage.

## 9. Managerial implications and conclusion

*Implications for system operators and planners..* Five lessons emerge for organisations that coordinate many operating units, from grid operators and regulators to international bodies that fund energy-transition programmes. First, the object to be learned is the *decision rule*, not the forecast: the cost-minimising combination of forecasting systems depends on the cost asymmetry each unit faces, and combining for accuracy and adding a margin leaves 2–3 points of the saving on the table in the United States and 8–10 points worldwide, with occasional extreme weeks. Second, planners need not choose between one rule for every system and a bespoke rule for each. A handful of data-driven segments, with each unit shrunk towards its segment’s rule, or even a single shared rule with unit-level shrinkage, captures most of the heterogeneity that matters, is estimated with far less noise than bespoke rules, and the grouped variant produces a segmentation that can be discussed with operators—who is grouped with whom, and why. Our evidence for the grouping itself is modest: it is statistically indistinguishable from simpler clustering rules, and decision-focused shrinkage comes close. Third, more flexible is not better: a neural mixture-of-experts gate with thousands of parameters, trained on the same decision loss, was seven percentage points more expensive worldwide than soft GDMA, much more expensive for systems with little history, and made 42% of the grid areas worse off than their own best candidate. Fourth, fairness in the sense of the criterion was nearly free: counting every area’s relative improvement equally produced a rule that was as cheap in total and halved the share of areas made worse off, whereas in storage siting a maximin rule protected the worst-served continent at a price of fairness of 7–9%, which is not negligible. Fifth, forecasting and investment are linked. In a stylised model that credits storage only with the commitment errors it absorbs, a better commitment rule reduces what storage can add—the reference rule needs 34 GW of optimally sited batteries to match the cost that soft GDMA achieves without any—and a storage programme sited on today’s error profile loses 2–4% of its value; planners should treat both as an interaction to be studied with the other value streams of storage, not as a siting recommendation. Because GDMA only re-weights forecasts that operators already produce, it can be adopted without replacing existing forecasting systems.

*Who would adopt it..* The grouping needs someone who sees many units at once. Natural adopters are organisations that already coordinate several operators—a regional transmission organisation computing rules for its member balancing authorities, the regional

coordination centres of European transmission system operators, the market operator of a multi-region market such as AEMO—and forecasting-service providers or regulators that work with many utilities. The data requirement is each unit’s candidate commitments and realised demand over the estimation window, which such coordinators typically hold already; units that cannot share them could keep their data and exchange only the gradients of their own cost with respect to the shared weights, as in federated learning. The rules are auditable because each unit’s commitment remains a convex combination of commitments it already understands, and the segment memberships can be discussed with the operators concerned. Operators remain responsible for reliability; GDMA is a way to set the day-ahead commitment that feeds their existing processes, not a replacement for reserve procurement or unit commitment.

*Limitations and extensions..* Our coverage of Africa and parts of Asia is thin, because public hourly demand data for South Africa, India, Japan and South Korea could not be retrieved in our computing environment, and the Chinese provinces are observed in a single year; the audit shows that such data-poor systems benefit much less than data-rich ones, and extending the panel is a priority. The fairness analysis measures income at the level of countries and U.S. states; the distribution of benefits across the households served within an area would require data we do not have. Outside the United States the reference forecaster is a model forecast, because the official day-ahead forecasts of European transmission system operators and of AEMO were not part of the data we could retrieve; savings against operator practice in those systems remain to be established. Even in the United States the reference is a stylised policy, the official forecast plus a rolling quantile margin, not what operators do when they clear markets and procure reserves. The cost calibration is stylised: shortfall costs are tied to a proxy based on the variability of interchange, which is not transfer capability, rather than to market prices or the value of lost load, and the single-period newsvendor abstracts from ramping, network constraints, multi-day commitments and from the fact that demand served in a load-shedding event understates true demand. The weather candidates use archived GFS and GEFS forecasts at 0.5–1 degree resolution, only from late 2020, and only for the areas where the 00 UTC run is available before gate closure. With humidity, wind, radiation and the ensemble spread they roughly halve the gap to the operators’ forecasts during Winter Storm Elliott, but the combinations remain about 10% more expensive than the operators over those days even though one candidate, designed with that storm in view, is 7% cheaper on its own: weights learned from the previous weeks cannot anticipate a storm. Letting the combination weights respond to the ensemble spread, using calibrated ensemble post-processing, and longer archives such as the GEFS reforecast, are natural extensions. The storage stage takes the commitment rule as given and credits batteries only with the commitment errors they absorb; jointly optimising commitments and battery dispatch, and adding the arbitrage, adequacy and renewable-integration value of storage, would make the siting problem much richer, and could change where storage should go. The theory treats the number of segments as given and relies on a high-level concentration condition; a formal analysis of the hold-out choice of  $G$  and  $\kappa$ , and of strong cross-sectional dependence through common weather shocks, is left for future work.

*Conclusion..* Grouped decision-focused model averaging combines three ideas that have so far developed separately—optimal model averaging, decision-focused learning and latent group structures—into one estimator, with guarantees for the idealised version, and links it to a stylised siting problem for new capacity. With only the information available at

the day-ahead gate closure, it turns the forecasts operators already have into commitment rules that are 12% cheaper worldwide, and 13–17% cheaper in the United States, than committing on a reference forecaster; the gain comes mostly from deciding for the decision and from sharing structure across units, with grouping a smaller refinement. The limits are as informative as the gains: no rule beats the operators during an abrupt storm, entering units are poorly served by every rule, and the value of better commitment for storage planning depends on value streams that we do not model.

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The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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### **CRedit authorship contribution statement**

**Quang-Vinh Dang:** Conceptualization, Methodology, Software, Formal analysis, Data curation, Investigation, Visualization, Writing – original draft, Writing – review & editing.

### **Data availability**

All data used in this study are publicly available from the sources cited in Sections 6 and 7. The code that downloads the data, builds the panels and reproduces every table and figure is available at [https://github.com/vinhqdang/economics\\_data\\_science](https://github.com/vinhqdang/economics_data_science).

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## Appendix A. Proofs

Throughout,  $\|\cdot\|_1$  is the  $\ell_1$  norm and constants  $c, C$  may change from line to line.

*Covering the simplex.* For  $\epsilon \in (0, 1]$  there is a set  $\mathcal{N}_\epsilon \subset \Delta_M$  with  $|\mathcal{N}_\epsilon| \leq (3/\epsilon)^M$  such that every  $\mathbf{w} \in \Delta_M$  has a point  $\mathbf{w}' \in \mathcal{N}_\epsilon$  with  $\|\mathbf{w} - \mathbf{w}'\|_1 \leq \epsilon$  (volumetric argument for the  $\ell_1$  ball, e.g. Vershynin, 2018, Ch. 4). Hence  $\mathcal{N}_\epsilon^G$  is an  $\epsilon$ -net of  $\Delta_M^G$  in the norm  $\max_g \|\mathbf{w}_g - \mathbf{w}'_g\|_1$ , of cardinality at most  $(3/\epsilon)^{GM}$ .

*Lipschitz property.* For all  $(\gamma, \mathbf{W})$  and  $(\gamma, \mathbf{W}')$  with  $\max_g \|\mathbf{w}_g - \mathbf{w}'_g\|_1 \leq \epsilon$ , Assumption 1 gives  $|\hat{R}(\gamma, \mathbf{W}) - \hat{R}(\gamma, \mathbf{W}')| \leq \bar{a}B\epsilon$  and  $|R(\gamma, \mathbf{W}) - R(\gamma, \mathbf{W}')| \leq \bar{a}B\epsilon$ .

*Proof of Theorem 1*

Fix  $\epsilon = 1/(NT_e)$ . The set  $\mathcal{F} = \{1, \dots, G\}^N \times \mathcal{N}_\epsilon^G$  has cardinality at most  $G^N(3NT_e)^{GM}$ . By Assumption 2 and the union bound,

$$\Pr \left( \max_{(\gamma, \mathbf{W}) \in \mathcal{F}} |\hat{R}(\gamma, \mathbf{W}) - R(\gamma, \mathbf{W})| > x \right) \leq 2G^N(3NT_e)^{GM} \exp(-cNT_e x^2/L^2).$$

Setting the right-hand side equal to  $\delta$  gives, with probability at least  $1 - \delta$ ,  $\max_{\mathcal{F}} |\hat{R} - R| \leq x_\delta := L\{(GM \log(3NT_e) + N \log G + \log(2/\delta))/(cNT_e)\}^{1/2}$ . For an arbitrary  $(\gamma, \mathbf{W}) \in \mathcal{P}_G$  let  $\mathbf{W}'$  be its nearest net point; by the Lipschitz property,

$$\sup_{\mathcal{P}_G} |\hat{R} - R| \leq x_\delta + 2\bar{a}B\epsilon.$$

Let  $(\gamma^*, \mathbf{W}^*)$  satisfy  $R(\gamma^*, \mathbf{W}^*) \leq R_G^* + \eta$  for arbitrary  $\eta > 0$ . Since  $(\hat{\gamma}, \widehat{\mathbf{W}})$  minimises  $\hat{R}$ ,

$$R(\hat{\gamma}, \widehat{\mathbf{W}}) \leq \hat{R}(\hat{\gamma}, \widehat{\mathbf{W}}) + x_\delta + 2\bar{a}B\epsilon \leq \hat{R}(\gamma^*, \mathbf{W}^*) + x_\delta + 2\bar{a}B\epsilon \leq R_G^* + \eta + 2x_\delta + 4\bar{a}B\epsilon.$$

Letting  $\eta \downarrow 0$  and substituting  $\epsilon = 1/(NT_e)$  gives (5).  $\square$

*Proof of Corollary 1*

Theorem 1 with  $\delta = \delta_{N,T} \rightarrow 0$  slowly (for instance  $\log(1/\delta) = \log(NT_e)$ ) shows  $R(\hat{\gamma}, \widehat{\mathbf{W}}) - R_G^* = O_p(r_{N,T})$  with  $r_{N,T} = \{GM \log(NT_e)/(NT_e) + \log G/T_e + \log(NT_e)/(NT_e)\}^{1/2} \rightarrow 0$ . Dividing by  $R_G^* \geq \kappa$  yields  $R(\hat{\gamma}, \widehat{\mathbf{W}})/R_G^* - 1 = O_p(r_{N,T}/\kappa) \rightarrow_p 0$ . Because  $R(\hat{\gamma}, \widehat{\mathbf{W}}) \geq R_N^*$  always and  $R_G^* \geq R_N^*$ , the second claim follows once we show  $R_G^* = R_N^*$  under the stated structure. Let  $\mathbf{w}_g^0$  be a common minimiser of  $R_i$  for all  $i$  with  $\gamma_i^0 = g$  (padding with arbitrary vectors if  $G_0 < G$ ); then  $R(\gamma^0, \mathbf{W}^0) = N^{-1} \sum_i \min_{\mathbf{w}} R_i(\mathbf{w}) = R_N^*$ , so  $R_G^* \leq R_N^*$ , and the reverse inequality holds because  $\mathcal{P}_G$  is a subset of the unit-specific class.  $\square$

*Proof of Theorem 2*

*Step 1 (average consistency).* Under Assumption 3(a),  $R_G^* = R(\gamma^0, \mathbf{W}^0) = R_N^*$  and

$$\frac{\lambda}{N} \sum_{i=1}^N \left\| \widehat{\mathbf{w}}_{\hat{\gamma}_i} - \mathbf{w}_{\gamma_i^0}^0 \right\|_2^2 \leq R(\hat{\gamma}, \widehat{\mathbf{W}}) - R_G^* = o_p(1)$$

by Theorem 1, since  $\log N/T_e \rightarrow 0$  implies  $\log G/T_e \rightarrow 0$  and  $GM \log(NT_e)/(NT_e) \rightarrow 0$  for fixed  $G$  and  $M$ .

*Step 2 (consistency of the segment weights).* Fix  $\eta \in (0, d/4)$ . By Step 1 and Markov's inequality, the fraction of units with  $\|\widehat{\mathbf{w}}_{\hat{\gamma}_i} - \mathbf{w}_{\gamma_i^0}^0\|_2 > \eta$  is  $o_p(1)$ . Since every true segment  $g$  contains at least  $\pi N$  units, with probability approaching one each  $g$  contains a unit  $i$  with  $\|\widehat{\mathbf{w}}_{\hat{\gamma}_i} - \mathbf{w}_g^0\|_2 \leq \eta$ ; call  $h(g) = \hat{\gamma}_i$ . If  $h(g) = h(g')$  for  $g \neq g'$ , the triangle inequality gives  $\|\mathbf{w}_g^0 - \mathbf{w}_{g'}^0\|_2 \leq 2\eta < d$ , contradicting Assumption 3(b). Thus  $h$  is injective, hence a bijection of  $\{1, \dots, G\}$ , and after relabelling  $\max_g \|\widehat{\mathbf{w}}_g - \mathbf{w}_g^0\|_2 \leq \eta$  with probability approaching one. Since  $\eta$  is arbitrary, (i) follows.

*Step 3 (memberships).* On the event  $E_\eta = \{\max_g \|\widehat{\mathbf{w}}_g - \mathbf{w}_g^0\|_2 \leq \eta\}$ , unit  $i$  with  $\gamma_i^0 = g$  can be assigned to  $h \neq g$  only if  $\widehat{R}_i(\widehat{\mathbf{w}}_h) \leq \widehat{R}_i(\widehat{\mathbf{w}}_g)$ ; every global minimiser assigns each unit to a segment minimising its own empirical cost, since otherwise reassigning the unit would lower (2). By Assumption 3 and the Lipschitz property (with  $\|\cdot\|_1 \leq \sqrt{M}\|\cdot\|_2$ ),

$$R_i(\widehat{\mathbf{w}}_h) - R_i(\widehat{\mathbf{w}}_g) \geq \lambda(d - \eta)^2 - \bar{a}B\sqrt{M}\eta \geq \lambda d^2/4 =: \Delta$$

for  $\eta$  small enough. Hence misclassification of unit  $i$  on  $E_\eta$  requires  $\sup_{\mathbf{w} \in \Delta_M} |\widehat{R}_i(\mathbf{w}) - R_i(\mathbf{w})| \geq \Delta/2$ . Covering  $\Delta_M$  by  $\mathcal{N}_\epsilon$  with  $\epsilon = \Delta/(8\bar{a}B)$  and applying the unit-level part of Assumption 2,

$$\Pr\left(\sup_{\mathbf{w}} |\widehat{R}_i(\mathbf{w}) - R_i(\mathbf{w})| \geq \Delta/2\right) \leq 2\left(\frac{24\bar{a}B}{\Delta}\right)^M \exp\left(-\frac{cT_e\Delta^2}{16L^2}\right) =: C_M e^{-c'T_e}.$$

Because the supremum is over all  $\mathbf{w}$ , the bound does not require independence between  $\widehat{\mathbf{W}}$  and unit  $i$ 's data. A union bound over units gives  $\Pr(\widehat{\gamma} \neq \gamma^0) \leq \Pr(E_\eta^c) + NC_M e^{-c'T_e}$ , which tends to zero because  $\log N/T_e \rightarrow 0$ . This is (ii).

*Step 4 (oracle equivalence).* On the event  $\{\widehat{\gamma} = \gamma^0\}$ ,  $\widehat{\mathbf{W}}$  minimises  $\widehat{R}(\gamma^0, \cdot)$ , so  $\widehat{\mathbf{W}} = \widetilde{\mathbf{W}}$  (if the minimiser is not unique, the two estimators select from the same set). By (ii) this event has probability approaching one.  $\square$

## Appendix B. Out-of-sample risk

The theory concerns the population risk  $R$  evaluated at the estimated policy. In a rolling application the policy estimated on periods  $1, \dots, T$  is applied in periods  $T + 1, \dots, T + k$ . If  $\{\mathbf{z}_{it}\}$  is  $\beta$ -mixing with coefficients  $\beta(\cdot)$ , a coupling argument (Yu, 1994) shows that the expected cost of the estimated policy in period  $T + s$  differs from  $\mathbb{E} R(\widehat{\gamma}, \widehat{\mathbf{W}})$  by at most  $L\beta(s)$ . The bounds of Section 4 therefore also bound the out-of-sample cost up to a term that vanishes with the gap between estimation and application.

## Appendix C. Balancing authorities

## Appendix D. Additional results for the global panel

Table C.20: The 46 balancing authorities, ordered by mean hourly demand over 2019–2026. Flex.: interchange flexibility in 2019 (range between the 5th and 95th percentiles of hourly total interchange, divided by mean demand).  $\tau_i$ : calibrated critical ratio.

| Code | Balancing authority                                                | Mean MW | Flex. | $\tau_i$ |
|------|--------------------------------------------------------------------|---------|-------|----------|
| PJM  | PJM Interconnection, LLC                                           | 92,507  | 0.06  | 0.947    |
| MISO | Midcontinent Independent System Operator, Inc.                     | 74,189  | 0.07  | 0.943    |
| ERCO | Electric Reliability Council of Texas, Inc.                        | 49,750  | 0.03  | 0.950    |
| SWPP | Southwest Power Pool                                               | 32,259  | 0.08  | 0.940    |
| SOCO | Southern Company Services, Inc. - Trans                            | 27,075  | 0.12  | 0.927    |
| CISO | California Independent System Operator                             | 25,319  | 0.29  | 0.900    |
| TVA  | Tennessee Valley Authority                                         | 18,637  | 0.09  | 0.937    |
| NYIS | New York Independent System Operator                               | 17,306  | 0.18  | 0.910    |
| FPL  | Florida Power & Light Co.                                          | 15,873  | 0.10  | 0.933    |
| ISNE | ISO New England                                                    | 13,228  | 0.16  | 0.920    |
| DUK  | Duke Energy Carolinas                                              | 12,212  | 0.23  | 0.907    |
| CPLE | Duke Energy Progress East                                          | 7,045   | 0.37  | 0.880    |
| BPAT | Bonneville Power Administration                                    | 6,631   | 0.84  | 0.827    |
| PACE | PacifiCorp East                                                    | 5,803   | 0.42  | 0.877    |
| PSCO | Public Service Company of Colorado                                 | 5,312   | 0.11  | 0.930    |
| NEVP | Nevada Power Company                                               | 4,523   | 0.33  | 0.883    |
| LGEE | Louisville Gas and Electric Company and Kentucky Utilities Company | 4,166   | 0.17  | 0.913    |
| AZPS | Arizona Public Service Company                                     | 3,893   | 0.67  | 0.843    |
| SC   | South Carolina Public Service Authority                            | 2,901   | 0.32  | 0.890    |
| LDWP | Los Angeles Department of Water and Power                          | 2,867   | 0.30  | 0.897    |
| SCEG | Dominion Energy South Carolina, Inc.                               | 2,814   | 0.15  | 0.923    |
| AECI | Associated Electric Cooperative, Inc.                              | 2,658   | 0.52  | 0.867    |
| TEC  | Tampa Electric Company                                             | 2,530   | 0.26  | 0.903    |
| PGE  | Portland General Electric Company                                  | 2,523   | 0.53  | 0.863    |
| PACW | PacifiCorp West                                                    | 2,392   | 0.55  | 0.860    |
| FMPP | Florida Municipal Power Pool                                       | 2,135   | 0.17  | 0.917    |
| IPCO | Idaho Power Company                                                | 2,097   | 0.79  | 0.833    |
| BANC | Balancing Authority of Northern California                         | 1,993   | 0.59  | 0.853    |
| PNM  | Public Service Company of New Mexico                               | 1,645   | 0.64  | 0.847    |
| TEPC | Tucson Electric Power                                              | 1,591   | 0.71  | 0.837    |
| JEA  | JEA                                                                | 1,532   | 0.32  | 0.893    |
| AVA  | Avista Corporation                                                 | 1,460   | 0.70  | 0.840    |
| NWMT | NorthWestern Corporation                                           | 1,358   | 1.07  | 0.810    |
| SCL  | Seattle City Light                                                 | 1,076   | 0.92  | 0.817    |
| WALC | Western Area Power Administration - Desert Southwest Region        | 1,045   | 0.95  | 0.813    |
| EPE  | El Paso Electric Company                                           | 1,029   | 0.46  | 0.873    |
| GCPD | Public Utility District No. 2 of Grant County, Washington          | 696     | 1.58  | 0.807    |
| CPLW | Duke Energy Progress West                                          | 564     | 0.61  | 0.850    |
| TPWR | City of Tacoma, Department of Public Utilities, Light Division     | 540     | 0.56  | 0.857    |
| IID  | Imperial Irrigation District                                       | 441     | 0.90  | 0.820    |
| TIDC | Turlock Irrigation District                                        | 328     | 0.81  | 0.830    |
| TAL  | City of Tallahassee                                                | 322     | 0.32  | 0.887    |
| DOPD | PUD No. 1 of Douglas County                                        | 245     | 2.21  | 0.803    |
| GVL  | Gainesville Regional Utilities                                     | 240     | 0.50  | 0.870    |
| CHPD | Public Utility District No. 1 of Chelan County                     | 218     | 4.71  | 0.800    |
| HST  | City of Homestead                                                  | 71      | 0.89  | 0.823    |

Table D.21: Global panel without the weather candidate (eight candidates): system cost relative to the single best forecaster for estimation windows of 14 and 28 days (world).

| Method                            | $W = 14$ | $W = 28$ |
|-----------------------------------|----------|----------|
| Reference forecaster <sup>a</sup> | 1.000    | 1.000    |
| Equal weights                     | 1.315    | 1.380    |
| Best single (per area)            | 0.887    | 0.942    |
| Pooled DF weights                 | 0.874    | 0.929    |
| Per-area DF weights               | 0.857    | 0.918    |
| Shrinkage to pooled               | 0.846    | 0.911    |
| Two-step $k$ -means               | 0.843    | 0.904    |
| Quantile-regression comb.         | –        | 1.038    |
| Forecast-then-commit, per area    | 0.973    | 1.011    |
| Forecast-then-commit, grouped     | 0.948    | 0.988    |
| Neural gate (deep MoE)            | 0.918    | 0.968    |
| GDMA                              | 0.849    | 0.905    |
| <b>Soft GDMA</b>                  | 0.841    | 0.906    |
| Fair soft GDMA                    | –        | 0.905    |

Table D.22: Global panel without the weather candidate and with a common critical ratio  $\tau = 0.9$  for every grid area ( $W = 28$ ): system cost relative to the single best forecaster. The neural gate and the fair variant were not run for this specification.

| Method                            | World        | North America | Europe       | Oceania      | Asia         | Africa       |
|-----------------------------------|--------------|---------------|--------------|--------------|--------------|--------------|
| Reference forecaster <sup>a</sup> | 1.000        | 1.000         | 1.000        | 1.000        | 1.000        | 1.000        |
| Equal weights                     | 1.285        | 1.434         | 1.171        | 1.027        | 1.164        | 1.032        |
| Best single (per area)            | 0.862        | 0.834         | 0.889        | 0.860        | 0.888        | 0.878        |
| Pooled DF weights                 | 0.865        | 0.834         | 0.893        | 0.920        | 0.864        | 0.828        |
| Per-area DF weights               | 0.832        | 0.788         | 0.869        | 0.856        | 0.886        | 0.839        |
| Shrinkage to pooled               | 0.828        | 0.787         | 0.863        | 0.865        | 0.857        | 0.817        |
| Two-step $k$ -means               | 0.826        | 0.793         | 0.853        | 0.858        | 0.857        | 0.837        |
| Forecast-then-commit, per area    | 0.894        | 0.887         | 0.880        | 0.854        | 1.039        | 0.846        |
| Forecast-then-commit, grouped     | 0.888        | 0.903         | 0.855        | <b>0.853</b> | 1.006        | 0.819        |
| GDMA                              | 0.828        | 0.796         | 0.854        | 0.868        | 0.859        | 0.812        |
| <b>Soft GDMA</b>                  | <b>0.822</b> | <b>0.786</b>  | <b>0.853</b> | 0.859        | <b>0.857</b> | <b>0.810</b> |